

Bottom-Coding for the Measurement of Global Poverty and Inequality

Nishant Yonzan
Minh C. Nguyen
Christoph Lakner
Aart Kraay
Dean Jolliffe
Haoyu Wu
Gabriel Lara Ibarra



WORLD BANK GROUP

Development Data Group

Prosperity Vertical &

Fiscal Policy and Growth Global Department

April 2026



Reproducible Research Repository

A verified reproducibility package for this paper is available at <http://reproducibility.worldbank.org>, click **here** for direct access.

Abstract

Distribution sensitive welfare and poverty measures are designed to respond to low values of individual income or consumption. However, very small positive values can have an extreme influence on many distribution sensitive measures, and some measures are not defined for nonpositive values. Although income or consumption is often truncated or bottom-coded at some strictly positive threshold to avoid these difficulties, it is unclear where the threshold should be drawn and data providers follow different

practices. This paper systematically explores various options for bottom-coding consumption and income distributions when calculating summary welfare and poverty measures for cross-country and over-time comparisons. Using more than 800 consumption surveys from across the world and employing various methodologies, the paper suggests that \$0.25 per person per day in 2017 purchasing power parity is a suitable bottom-coding threshold in this setting.

This paper is a product of the Development Data Group, Development Economics, the Office of the Chief Economist, Prosperity Vertical, and the Fiscal Policy and Growth Global Department. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at <http://www.worldbank.org/prwp>. The authors may be contacted at nyonzan@worldbank.org. A verified reproducibility package for this paper is available at <http://reproducibility.worldbank.org>, click [here](#) for direct access.



The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

Bottom-Coding for the Measurement of Global Poverty and Inequality

Nishant Yonzan  Minh C. Nguyen  Christoph Lakner  Aart Kraay

 Dean Jolliffe  Haoyu Wu  Gabriel Lara Ibarra¹

Authorized for distribution by Indermit Gill, Chief Economist & Senior Vice President, Development Economics, World Bank Group

Keywords: Bottom code, consumption floor, measurement errors, distribution-sensitive index, parametric tail

JEL Classifications: C1, D60, I32

¹ All authors are with the World Bank. Corresponding author: Nishant Yonzan (nyonzan@worldbank.org). We would like to thank the Global Poverty Working Group of the World Bank and the participants of IARIW-World Bank-VNU/UEB conference for helpful comments. The names of the authors are in random order, using the author randomization tool of the American Economic Association, which is archived [here](#). We acknowledge partial financial support from the Foreign, Commonwealth & Development Office of the UK government through the Data and Evidence for Tackling Extreme Poverty (DEEP) Research Program. The views expressed here are the authors', and do not necessarily reflect the official views of the World Bank, its Executive Directors, or the countries they represent.

1. Introduction

Many distribution sensitive poverty and welfare measures, as well as inequality measures, are designed to respond strongly to low values of individual consumption or income. However, it is well known that this desired sensitivity also implies that very small values of consumption or income can be highly influential when making comparisons based on distribution sensitive indices (Cowell and Victoria-Feser, 1996; Cowell and Flachaire, 2007). Consider for example a standard

Atkinson welfare index $A_\varepsilon(c) = \left(\frac{1}{N} \sum_{i=1}^N c_i^{1-\varepsilon}\right)^{\frac{1}{1-\varepsilon}}$, where $c = (c_1, \dots, c_N) \in R_{>0}^N$ represents the distribution of consumption across N individuals in a society. When the coefficient of inequality aversion ε is greater than one, the Atkinson index approaches zero when the consumption of just one person approaches zero, irrespective of the consumption of everyone else in society.

These concerns about potentially excessive sensitivity of welfare comparisons to low measured values of individual consumption are not hypothetical. Consider the case of Benin in 2015. Average consumption per person per day is \$3, but 3 percent of observations report extremely low, below \$0.25, daily per capita consumption.² Using all observations, the Atkinson index with $\varepsilon = 2$ for Benin is equal to \$1.09. Dropping just 3 percent of observations with very low consumption results in a sharp increase in the Atkinson index to \$1.40. Recalling the equally distributed equivalent income interpretation of the Atkinson index, this increase in welfare is equivalent to a $\$1.40/\$1.09 \times 100 = 29$ percent increase in consumption *for everyone in Benin*. Although many users may in general want to base poverty and welfare comparisons on distribution-sensitive measures, they may be uncomfortable with the extreme sensitivity of these measures to a small count of presumably implausibly low values of consumption.

Concerns about sensitivity are compounded by the presumption that near-zero values of observed consumption likely reflect error in respondent recall (Ravallion, 2016), enumerator recording, or item (i.e., partial) nonresponse. For example, 13 consumption surveys in the World Bank's Global Monitoring Database have some observations with zero monthly reported consumption (i.e., implausible values).³ Very low or zero reported consumption is likely to be due to measurement error, broadly defined, given that there is a biological minimum consumption level required to sustain life (Herforth et al., 2022). In addition, to the extent that low consumption is correlated with low numeracy and literacy, surveys that require respondents to quantify consumption of several items (and in some cases, convert quantities from nonstandard to standardized units) over a recall period covering several days may be less accurate for

² All monetary figures in this paper are expressed in constant 2017 US dollars using the 2017 purchasing power parities. The World Bank has recently adopted the 2021 PPPs in its global poverty monitoring. Based on the average conversion factor between 2017 and 2021 PPPs (Foster et al., 2025), \$0.25 at \$2017 PPP corresponds to \$0.28 at \$2021 PPP.

³ The Global Monitoring Database (GMD) is a repository of survey microdata maintained by the World Bank, primarily for the purposes of reporting on global poverty. The GMD has more than 2,500 country-year surveys spanning roughly half a century. Summary statistics on these distributions are reported in the World Bank's Poverty and Inequality Platform (PIP). See Section 2 for more details.

households with low consumption levels. Even if measurement error is not more prevalent for low consumption households, it will nonetheless be the case that mean-zero measurement error for households in extreme poverty will result in some observed values that fall below minimum levels required to sustain life.

In an era of declining trust in official statistics and data, the analysis and documentation of the sensitivity of key poverty and inequality statistics is a critical step to improve trust. International data providers typically bottom-code distributions at some threshold (i.e., replacing observations below the threshold with the threshold value), or truncate distributions by dropping observations below the threshold entirely. However, these decisions are not always well reported and there is no clear consensus on what this threshold should be -- different organizations follow different practices. For example, the Luxembourg Income Study bottom- (and top-) codes the distribution of log income at three times the inter-quartile range below (above) the first (third) quartile when reporting inequality measures (Neuenschwander, 2020). To give another example, until recently, all indicators in the World Bank's Poverty and Inequality Platform (PIP) were calculated using a distribution of income or consumption that is truncated at zero, and further adjustments were made for those indicators that mathematically cannot accommodate zero values.⁴

Moreover, precisely because many distribution sensitive measures are highly sensitive to low values of income or consumption, they can also be highly sensitive to the bottom-coding or truncation threshold when this threshold also is low. Returning to our Benin example, truncating daily per capita consumption at a lower value of \$0.15 results in an Atkinson index of \$1.27, while truncating at a higher value \$0.50 results in an Atkinson index of \$1.97. More generally, there is an uncomfortable tradeoff: if the threshold is set too low, distribution-sensitive welfare and poverty measures may be overly sensitive to the value of the threshold, while if the threshold is set too high, the measures may not adequately reflect the well-being of a significant number of the poorest individuals in a society (if the distribution is bottom-coded) or ignore them entirely (if the distribution is truncated).

Given the arbitrariness of existing approaches and the potential sensitivity of welfare, inequality and poverty comparisons to alternative truncation or bottom-coding thresholds, there is value in systematically exploring options for these thresholds and their implications for comparisons. We do this using poverty, inequality and welfare measures reported in the PIP for 1,826 household surveys. This setting is particularly relevant given the increasing focus on distribution sensitive measures beyond the traditional poverty headcount measure, which only counts the number of people below and thus is unaffected by very low or negative values (World Bank, 2023).

We focus on consumption surveys because simple biology dictates that average per capita consumption over a non-trivial time interval must be strictly positive. We then apply a battery of

⁴ For instance, in the case of the mean log deviation, observations with a value of zero were replaced with one (giving these observations a difference value of $\log(\text{mean}) - \log(1)$), while those observations were dropped in the case of the Watts index. As of the September 2024, the bottom-coding threshold proposed in this paper has been applied to all estimates in PIP.

statistical methods designed to estimate the non-zero minimum value for consumption that is implied by viewing the observed data through a variety of statistical lenses. For example, we assume that the observed data is generated by several different parametric probability distribution functions with strictly positive support, and we estimate the values of the lower bound on consumption implied by the observed data and the corresponding distributional assumptions. We compare these statistical estimates with existing evidence on the cost of the least-cost energy sufficient diet, which is a very frugal diet that only guarantees a minimum amount of calories without providing other nutrients that are needed for a healthy life.

These approaches yield a range of possible values for a threshold. To narrow down options within this range, we also compare the contribution of the bottom decile to welfare measures calculated with bottom-coded data with the corresponding contribution of the bottom decile when welfare measures are calculated in hypothetical well-behaved distributions, for a variety of values for the bottom-coding threshold. In this discussion and in the remainder of the paper, we focus on bottom-coding as opposed to truncation, because bottom-coding preserves some information on the consumption of the poorest individuals, while truncation discards this information completely.

Triangulating across these different approaches, we propose a bottom-coding threshold of \$0.25 per person per day for consumption. This threshold is close to the minimum across countries of the estimated cost of the least-cost energy sufficient diet noted above. This also is a very low value. For over 75 percent of household consumption surveys in the PIP after 2000, there are no observations at all with consumption below \$0.25, so bottom-coding at this threshold has no impact whatsoever on poverty, inequality and welfare measures. For another 22 percent of surveys, less than 1 percent of the population in the survey have consumption below \$0.25. Among the rest (17 surveys), only 6 surveys have more than 2 percent of the population below \$0.25. This means that even in these 17 surveys, bottom-coding at \$0.25 has only limited impact on welfare measures. For example, the average change in the Atkinson(2) index in these 17 surveys is less than 7 percent, and the largest change in the Atkinson(2) index is 21 percent for Lesotho's 2002 survey where 6 percent of observations fall below the bottom-coding threshold.

Our primary focus in this paper is to identify a suitable bottom-coding threshold for consumption distributions. Bottom-coding income distributions poses an additional conceptual challenge: measured income over a given period can legitimately be zero or negative when people can temporarily finance non-zero consumption during this period by drawing down savings. In addition to the purely mathematical problem that some distribution sensitive poverty, inequality and welfare measures cannot accommodate negative incomes, this introduces a new type of measurement error: it is difficult to distinguish between individuals with persistently low income, consumption and welfare, and well-off individuals with sufficiently high savings that they can easily maintain their consumption and living standards when income temporarily is very low or even negative. Since existing evidence suggests that households with negative incomes are relatively well-off (Hlasny, Ceriani, Verme, 2022), replacing very low or negative observed income

with a low bottom-coding threshold therefore risks under-estimating the well-being of these households, and therefore under-estimating inequality and distribution sensitive welfare measures.⁵ Furthermore, there is evidence to suggest that, especially for households with few resources, there are larger measurement errors in income than in consumption data (Meyer and Sullivan, 2003), making defining a threshold in income data much more challenging.

The appropriate treatment of very low income levels is just a subset of a much broader set of issues relating to the comparability of consumption and income distributions and the conceptual suitability of income as a measure of welfare at the individual level, which we do not address in this paper. Instead, we simply apply the proposed bottom-coding threshold of \$0.25 per person per day to the non-negative values of the income distributions to illustrate the sensitivity of resulting aggregate welfare measures to this threshold, recognizing that bottom-coded income may be a poor proxy for the actual well-being of many bottom-coded individuals. Less bottom-sensitive measures, such as the Gini index, are minimally affected by the \$0.25 per day bottom-code. The average change in the Gini index for the 10 consumption (income) surveys with the largest share of the population below the bottom-code is 0.7 (0.2) percent. Impact for the bottom-sensitive measures, such as the Atkinson inequality index with $\varepsilon = 2$, is limited to a handful of surveys, most of which are older than 20 years. Among the 10 consumption (income) surveys with the largest share of the population below \$0.25 per day, the average impact for the Atkinson inequality index with $\varepsilon = 2$ is 12 percent (7 percent).

The rest of this paper proceeds as follows. Section 2 describes the data, and Sections 3 and 4 introduce the various methods used for estimating a bottom-coding threshold and present the results. Sections 5 and 6 summarize the various approaches and report the impact of the thresholds on various poverty and inequality measures. Section 7 concludes.

⁵ We have examined 20 income surveys in the World Bank's Global Monitoring Database (see Section 2) in which measured incomes are zero in the lowest percentiles of the income distribution, *and* a consumption survey is available in the same year. These countries are all upper middle-income or high-income countries in Eastern Europe. Pooling across all surveys, median consumption over those percentiles for which reported incomes are zero is \$2.64 per day, with the lowest consumption percentile being \$0.84 per day. For an additional 14 surveys spanning 4 countries, information on both consumption and income is available in the same survey. For Nicaragua, Haiti, and St. Lucia, countries that have reported income of zero, the minimum consumption spans between \$0.14 and \$1.93.

2. Data

The primary data source for this study is the **Global Monitoring Database (GMD)**, a comprehensive repository of survey microdata maintained by the World Bank. The GMD consists of nationally representative household surveys conducted by national statistical offices, sometimes in collaboration with the World Bank. These surveys are subsequently compiled, processed, and harmonized to ensure a minimum level of consistency and comparability across different countries and time periods. From the GMD, we obtained data for 122 countries covering the period 1981 to 2023. For an additional **34 countries we use microdata from the European Union's Statistics** on Income and Living Condition surveys (EU-SILC) covering the period 2004 to 2023.⁶

The household surveys in the GMD provide extensive information on either household income or household consumption. We refer to these as “income” or “consumption” surveys, and we sometimes use the generic term “welfare aggregate” to refer to both when the distinction is not relevant. In addition, the GMD contains various harmonized demographic variables, such as household size, age, gender, and the rural/urban classification of households. The GMD serves as the primary data source for the World Bank’s poverty and inequality estimates, as published in the Poverty and Inequality Platform (PIP), as well as reported in the UN’s SDG database.

This paper uses 1,826 household surveys (country-year observations) conducted in 156 economies. Among them, 795 are consumption surveys, 1,003 are income surveys, and 14 report both income and consumption. Table 1 provides an overview of the distribution of surveys conducted across different regions, distinguishing between consumption and income surveys. There is significant regional and temporal variation in the availability of consumption and income surveys, as well as the frequency of surveys by country. Notably, Europe & Central Asia (ECA), which includes the advanced economies of Western Europe, has most surveys available (921 surveys, split between 372 consumption surveys and 549 income surveys). In contrast, most surveys in Latin America & the Caribbean (LAC) have income as the welfare aggregate, while most surveys in Sub-Saharan Africa (SSA) have consumption. ECA has 20 surveys available per country on average, compared with only 4 surveys per country in SSA. Over time, data availability has increased strongly, with the post-2000 period showing a peak in the number of surveys available, with more than 40 new surveys becoming available in some years (Figure 1).

⁶ GMD also hosts an additional 473 country-year observations that are based on grouped data (e.g., percentile groups published by NSOs or bins extracted from the Luxembourg Income Study (LIS) database). Only surveys with microdata at the individual or household level are included in this study. At the regional level, North America is not covered in the data used in this paper, since the surveys that the GMD includes from this region are based on LIS.

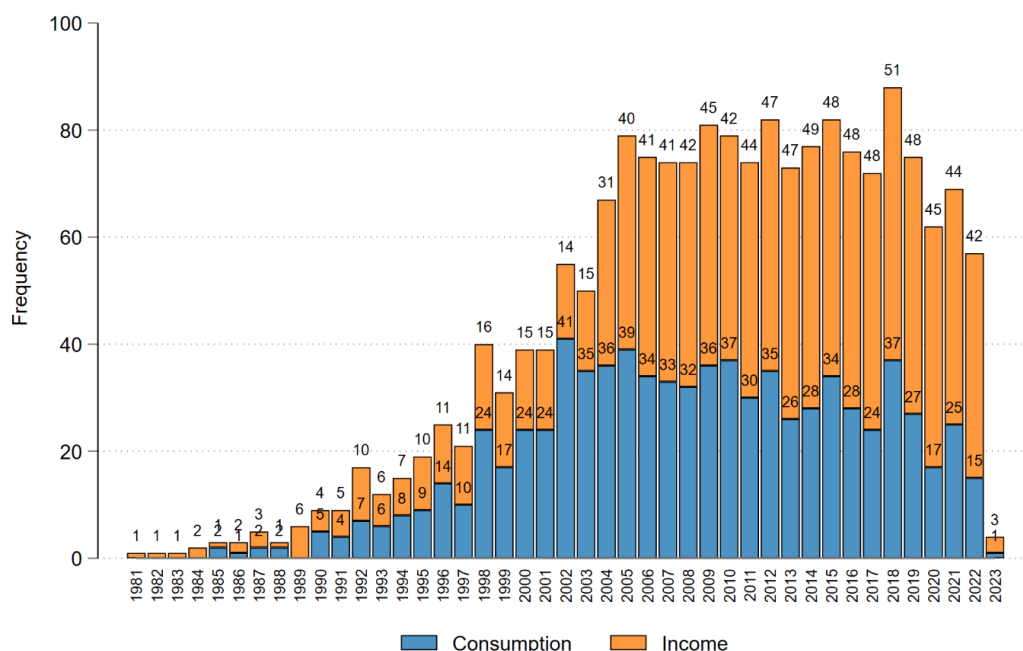
Table 1: Distribution of available survey data by region

Region	Number of surveys		Number of countries	Average survey per country
	Consumption	Income		
East Asia & Pacific	120	17	20	7
Europe & Central Asia	372	549	47	20
Latin America & the Caribbean	18	434	25	18
Middle East & North Africa	56	15	12	6
South Asia	45	0	7	6
Sub-Saharan Africa	198	2	45	4
Total	809	1017	156	12

Note: Regional groupings are based on World Bank FY 2025 classifications (World Bank, n.d.).

Across 1,826 surveys, negative values for household income were observed in 457 surveys. These are reported in Table 2. Among these surveys, 0.20 percent of the survey population had negative income on average, with a maximum share of 1.30 percent. Zero incomes are observed in 762 surveys, and zero consumption is observed in an additional 11 consumption surveys. Combining these, on average, 0.38 percent of the population had zero reported income or consumption, with a maximum share of 10.68 percent.

Figure 1: Frequency of available survey data by type of welfare aggregate used



Note: The chart shows the number of surveys available for each calendar year. The surveys are disaggregated by the type of welfare aggregate – income or consumption – used. Surveys conducted in a fiscal year are reported as a survey in the starting calendar year. EUSILC surveys collected in year T are reported in year T-1 as the income reference period refers to the previous calendar year.

Table 2: Prevalence of negative and zero observations

	Number of surveys		Share of population, %			
	All	Income	Mean	p75	p90	Max
Negative	457	457	0.20	0.28	0.46	1.30
Zero	773	762	0.38	0.31	1.10	10.68

Note: The table reports the (weighted) share of population and number of surveys influenced by negative or zero values. P75 and p90 correspond to the 75th and 90th percentiles, respectively.

3. Methodology

In this section, we describe four methods to identify plausible minimum values for consumption that could serve as a threshold to bottom-code consumption distributions. First, following Ravallion (2016), we estimate a “permanent consumption floor” (see also Jolliffe, Margitic, Ravallion 2024; Beduk, Jara, Ravallion, *forthcoming*). This approach can be thought of as fitting a smooth non-parametric distribution to the bottom tail of the income distribution while imposing the restriction that the bottom end of the range of the distribution is strictly positive. Second, we fit a variety of parametric distributions whose domains have a finite lower bound and examine whether the estimated lower bound is strictly positive and can serve as a plausible bottom-coding threshold. Third, we use an outlier identification methodology, such as that adopted by the Luxembourg Income Study (LIS) database (Neugschwender, 2020) and examine whether the threshold for small outliers is strictly positive and can serve as a plausible bottom-coding threshold. Fourth, we review existing evidence of the minimum cost of an energy-sufficient diet (Herforth et al., 2022) as an alternative justification for a biologically dictated minimum threshold for consumption.

We deploy these methods in consumption surveys where, as discussed above, there is a logical reason to focus on a strictly positive lower bound. However, we report the impact of choice of the bottom-coding threshold in both consumption and income surveys, to illustrate the sensitivity of income distributions to low values. As noted in the introduction, we acknowledge that for income distributions, bottom-coding is likely to understate the actual well-being of better-off households who can afford to smooth consumption during periods of low income.

3.1. Permanent consumption floor using the Ravallion (2016) method

Ravallion (2016) argued against using the lower bound of observed consumption or income in survey data as the permanent floor since “observed minimum values [in survey data] are subject to transient factors, and measurement errors.” Instead, he argues that “the observed current consumption equals the floor for permanent consumption at some point within a range of low

observed consumption”, and that the probability of the observed consumption value being the consumption floor declines as the observed consumption increases.

In other words, the permanent minimum consumption, or consumption floor, is given by the function: $E[y^{min}|x] = \sum_{i=1}^n \phi(x_i)x_i$, where y is permanent consumption, x is observed consumption, and $\phi(\cdot)$ is the relative weight assigned to each observed value i . Furthermore, with the assumption that the probability of finding a floor that is above the poverty line, z , is zero, a permanent consumption floor (y_f) can be estimated by

$$y_f \equiv E_\alpha[y^{min}|x] = z \times \left(1 - \frac{P_{\alpha+1}}{P_\alpha}\right) \quad (1)$$

where P is the Foster-Greer-Thorbeck (FGT) poverty index (Foster et al., 1984) and $\alpha \geq 1$ is the degree of curvature. For instance, when $\alpha = 1$, linear weights are assumed and $P_\alpha = P_1$ is the poverty gap index, and $P_{\alpha+1} = P_2$ is the squared poverty gap index. The higher the degree of curvature, the larger the weight is assigned to the values at the bottom of the distribution compared to those near the poverty line. To implement this method, two parameters need to be chosen: the poverty line z and the weights or curvature α .

Ravallion (2016) uses the International Poverty Line (*IPL*) as the poverty line and reports results for $\alpha = 1, 2$. In a similar application, Beduk, Jara, Ravallion, (forthcoming), using longitudinal data for Great Britain, argue that using the income or consumption of the person at the 10th percentile as the poverty line along with $\alpha = 1$ constitutes a reasonable option to estimate a permanent consumption floor.

In short, the permanent consumption floor estimated with this approach is a weighted average consumption of the observations below a certain poverty line. It is important to highlight that the objective of our paper is to identify a suitable minimum value to bottom code the distribution to calculate poverty and inequality indicators. Identifying the permanent consumption floor might tell us something about the ceiling for the bottom-coding value.

3.2. Minimum value using various parametric distributional assumptions

The approach outlined in Ravallion (2016) justifies the use of a weighted average at the bottom of the distribution by emphasizing measurement concerns at the bottom of the distribution. The approach is, however, not clear on the distributional assumption underlying the probability weights, α , nor the choice of the poverty line.

An alternate approach is to fit a known parametric distribution with a domain that has a finite lower bound. We consider two variants of this approach. In the first variant, we fit a Beta distribution to the lower tail of the distribution – an approach that is conceptually similar to fitting

a Pareto distribution to the upper tail of income distributions to correct for missing or censored data at the top (see Cowell and Victoria-Feser, 1996; Cowell and Flachaire, 2007; Jenkins et al., 2011). This approach is easily applied because the lower bound can be estimated given an assumed poverty line (below which the Beta distribution is fit) and by exploiting the relationship between moments of the Beta distribution and FGT indices. The second variant is to fit known distributions with domains that have a finite lower bound to the entire distribution of consumption. We implement this by fitting three-parameter log normal (TPLN) and Dagum III distributions, both of which have a lower bound parameter that can be estimated jointly with the other parameters of the distribution. In all these methods, and unlike in the non-parametric approach in Ravallion (2016), there is no guarantee that the estimated lower bound on the domain of the distribution will be positive nor that it will be larger than the smallest observation in the survey. In fact, for the TPLN and the Dagum III distributions, the lower bound will be strictly smaller than the smallest observation in the survey.

3.2.1. Beta distribution

We assume that consumption between the consumption floor and z follows a shifted and rescaled Beta distribution, i.e., $y = y_f + (z - y_f)x$ with $x \sim \text{Beta}(\eta, \beta)$. As long as z is below the modal value of the overall density of the distribution, it is reasonable to impose the further restrictions that $\eta > 1$ and $\beta = 1$. Together, this ensures that the density of y is monotonically increasing over the interval $[y_f, z]$.

Using moment conditions of the Beta distribution and relating it with the summary poverty statistics as Ravallion (2016), the floor can be estimated as

$$y_f = z \left(1 + \frac{P_0 P_1 P_2}{P_0^2 P_2 - 2 P_0 P_1^2} \right) \quad (2)$$

where P_α is the FGT poverty index as in equation 1. See Appendix A for a detailed derivation of equation 2.

3.2.2. The three-parameter log normal distribution

The three-parameter log normal (TPLN) distribution is a shifted log-normal distribution that includes a threshold parameter in addition to the two familiar parameters of the basic lognormal distribution (see Aristizabal, 2012; Cohen and Whitten, 1980; Cohen, 1951; Wicksell, 1917).

The density function of the TPLN can be written as

$$f(x; \gamma, \mu, \sigma^2) = (2\pi\sigma^2)^{-\frac{1}{2}}(x - \gamma)^{-1} \exp\left\{-\frac{[\ln(x - \gamma) - \mu]^2}{2\sigma^2}\right\},$$

where, $\sigma > 0$, $-\infty < \mu < \infty$, and $\gamma < x < \infty$. μ is the mean, σ^2 is the variance, and γ is the threshold parameter of interest. Following Cohen & Whitten (1980), we estimate γ using a local maximum likelihood estimator (LMLE, see Appendix B) and method of moments. The formula for the method of moments estimator (ME) in equation 3 below is replicated from Cohen & Whitten (1980).

$$\hat{\gamma} = \bar{x} - s \times (\omega - 1)^{-\frac{1}{2}} \quad (3)$$

where, $\bar{x}$ is the sample mean, s is the sample standard deviation, ω is the real root from $\omega^3 + 3\omega^2 - (a^2 + 4) = 0$, and a is the third standard moment of the sample. Cohen & Whitten (1980) recommend using a version of the ME when the third moment of the distribution is less than one and using a version of the LMLE otherwise.

3.2.3. The Dagum III distribution

Dagum distributions have been used to fit distributions of income and wealth (see Dagum 1980; Jenkins and Jantti, 2005; Kleiber 2007; Kleiber 2008). The Dagum distribution, and others such as the Singh-Maddala and Burr III/XII, are a family of distributions that can be nested within the four-parameter generalized beta distribution. The Dagum distribution in particular is well suited for fitting income and wealth distributions as it improves on the log-normal distribution by allowing for a thick right tail and improves on the Pareto distribution by permitting an interior mode (Kleiber 2008).

The Dagum III is the variant with strictly positive support. The density function of the Dagum III distribution is as follows

$$f(x; a, b, p) = ab^ap(x - x_0)^{-a-1}(1 + b^a(x - x_0)^{-a})^{-p-1},$$

where, $a, b, p > 0$ and $x_0 \leq x < \infty$. We use maximum likelihood to estimate the minimum value, x_0 .⁷

⁷ The floor, x_0 , can also be estimated using $\left\{b \left[\left(1 - \frac{1}{a}\right)^{\frac{1}{p}} - 1 \right]\right\}^{-\left(\frac{1}{a}\right)}$ once the a, b , and p have been estimated (see Kleiber 2008 for details). The maximum likelihood formulation is reported in Appendix C.

3.3. Outlier detection method

Finally, if measurement issues are the main concern, outlier detection methods can be helpful in identifying such observations. The traditional approach for this kind of outlier detection is to assume normality in the distribution and then define a threshold in the standard normal distribution using z-scores. Typically, the method boils down to defining the floor that is at some distance from a moment of the distribution as in

$$y_f = \mu - z_\alpha \times \sigma$$

where the threshold y_f is z standard deviations σ below the mean μ .

One such method is applied to the standard inequality estimates reported in the Luxembourg Income Study (LIS) database (Neugschwender, 2020). LIS defines a floor (and also a ceiling) at three times the inter quartile range (IQR) below (above) the first quartile (third quartile) of the distribution of log income. We supplement this by reporting a range of values considering floors that are two- to four-times the IQR below the first quartile. Note, however, that the floors that are derived using this method are a function of the IQR , so the larger the IQR the lower the estimated floor, all else equal. In other words, distributions that are more unequal will tend to have lower floors. The floors can be expressed as

$$y_f = \exp(Q_1 - k \times (Q_3 - Q_1)) \quad (3)$$

where Q_q is the q^{th} quartile of the log of consumption and $k \in 2,3,4$. In addition to the lognormal used by LIS, Belotti, Mancini, and Vecchi (2024) describe various other choices for transforming the distribution to approximate normality. The authors also provide additional distance parameters beyond the IQR . Floors estimated using these approaches are reported in the Appendix.

3.4. Minimum cost of energy efficient diet

Beyond statistical approaches, the food poverty literature could also shed some light on what a biological consumption floor could be. FAO et al. (2020) defines three reference diets when analyzing cost and affordability: energy sufficient diet, nutrient adequate diet, and healthy diet. The energy sufficient diet is the minimum cost to meet calorie requirements using the least-cost available and typically consists of basic starchy staple food items in each country. The nutrient adequate diet provides, besides adequate calories, relevant nutrient intake values of macro- and micronutrients. A healthy diet goes further and includes a more diverse intake of foods from several different food groups. While coarse, the energy sufficient diet could be used as another benchmark for the value of a consumption floor, as it is intended to establish a lower bound on the cost of short-term survival.

Given the focus of analyzing global welfare, we use estimates produced by Food Prices for Nutrition, also known as the Food Nutrition Project. The project is a multi-year collaboration between the FAO, World Bank and Tufts University to study the cost and affordability of healthy diets using data on availability and prices that are routinely collected around the world (Tufts University, n.d.).

As described in FAO et al. (2020), the cost of an energy sufficient diet is estimated as follows. First, caloric needs are anchored to a reference adult and set at 2,329 kcal/day.⁸ Second, retail prices of each food item available for purchase in each market are obtained from the World Bank's International Comparison Program (ICP) for internationally standardized items (one nationally representative price per item) for 2017. In particular, food prices for 680 foods in 170 countries are then expressed in 2017 purchasing power parity (PPP) dollars using ICP's 2017 PPP factors (World Bank, 2020). Finally, to calculate least-cost diets for the energy sufficient diets, linear programming is used to select foods in the quantities needed to minimize the cost subject to caloric constraints. This approach yields a lower bound on the short-term cost of meeting energy needs. It is not intended to reflect typical consumption patterns of any particular country and does not account for dietary quality. The purpose is to provide a consistent, comparable benchmark across countries. In this study, we use the results for the year 2017.

4. Results

4.1. Permanent consumption floor using the Ravallion (2016) method

The permanent consumption floor is a function of the poverty line z and the degree of curvature or weights α . Given these two parameters, a permanent consumption floor can be estimated for each survey. Figure 2 reports the distribution of such floors estimated for all consumption surveys using all observations below $z = \text{InternationalPovertyLine}$ (IPL)– i.e., \$2 .15 per person per day in 2017 PPPs – and with $\alpha = 1$ (i.e., equally weighted average). Table 3 reports the summary statistics of floors using other combinations of the two parameters. The estimated floors in Figure 2 range from \$0.48 per capita per day to \$2.13 per capita per day. The minimum value of the floor has increased slightly over the three decades, from \$0.48 to \$0.73. At low levels of mean consumption, the average consumption floor increases with the increase in the mean consumption of the survey; however, the floor on average is relatively flat after roughly \$5 of mean daily per capita consumption. The average value of the floor among all consumption surveys included in the analysis is \$1.40 and the median is \$1.41.

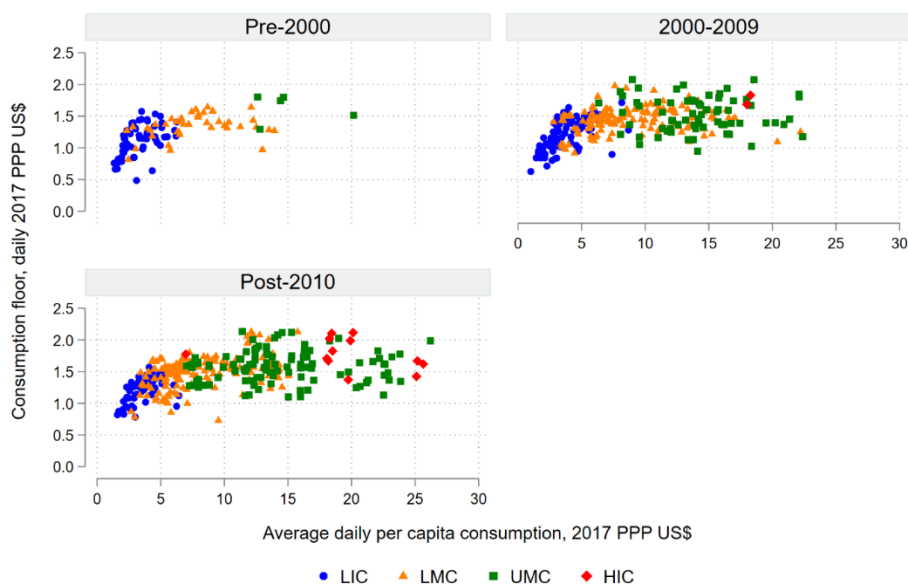
Using the *IPL* as the poverty line is somewhat arbitrary. Consider surveys where most of the population lives above the *IPL*. In those surveys, the floor would be defined by very few survey

⁸ The reference adult is a 30-year-old woman (57 kg, 1.63 m) with moderate physical activity. Using the same reference adult across countries helps ensure comparability. Furthermore, there is some evidence that this reference is similar to an age- and sex-population-weighted average.

observations if any. The opposite would be true for surveys that have most of the population living below the *IPL*. While not having enough observations might be limiting, there is at the same time no reason to believe that using all the observations in the survey improves the precision of the estimate of the floor.

To that end, various other thresholds are considered. A floor based on the national poverty line (*NPL*) might be preferred for a more nuanced treatment across all countries.⁹ However, the *NPL* is limited due to its lack of comparability across countries. For a more comparable measure, we consider the societal poverty line (*SPL*) or a given percentile of the consumption distribution. The *SPL* is a weakly relative poverty line defined by the World Bank (Jolliffe and Prydz, 2021), which is parameterized on the observed national poverty lines.¹⁰ Percentiles will more consistently capture an equal share of the population across all surveys to estimate the floor. Table 3 reports a range of estimates of the consumption floors using the national poverty line (*NPL*), and various other thresholds: the *SPL*, the 10th (*q10*) and 25th percentile (*q25*) of the consumption distribution.

Figure 2: Distribution of permanent consumption floors, $\alpha = 1$ and $z = IPL$



Note: The figure reports the distribution of permanent consumption floors for each consumption survey ($N = 809$) defined using a poverty line z equal to the international poverty line (*IPL*) – i.e., \$2.15 per person per day in 2017 PPPs – and a curvature parameter α equal to 1 (see equation 1 and Ravallion 2016). Surveys are grouped by the decade in which they were conducted. Consumption is reported in daily per capita 2017 PPP\$.

Acronyms: LIC – low-income country; LMIC – lower middle-income country; UMIC – upper middle-income country; HIC – high-income country. Income group classifications using definition for each calendar year.

⁹ National poverty lines in low, lower-middle, and most upper-middle income countries are absolute poverty lines, while most high-income countries use a relative poverty threshold (for additional detail, see Jolliffe et al., 2025).

¹⁰ Using the 2017 PPPs, the *SPL* is equal to the maximum of either i) \$1.15 + 0.5 x median or ii) \$2.15, the international poverty line (Jolliffe et al., 2025).

Furthermore, the curvature of the fit could be non-linear, i.e., $\alpha > 1$. A floor can be estimated using higher values of α , which gives more importance to the smaller values of consumption compared to larger values. Table 2 also reports estimates of the floor with two choices for α . A priori, it is unclear what the optimal value of α should be. We restrict the range of options to $\alpha = \{1, 4\}$. Any value of α within these bounds will generate a floor that is within the range of floors reported in Table 3.

Across the range of values for α , the minimum value of the permanent consumption floor ranges from \$0.02 to \$0.48 when considering all surveys and between \$0.10 and \$0.63 when using more recent (post 2000) surveys. Not surprisingly, the floors estimated using higher poverty lines are larger compared to floors estimated with lower poverty lines. This is because the floor in the Ravallion (2016) approach is weighted average of consumption values below some poverty line; the higher the line, the higher the consumption values are incorporated in this average. For example, the median floor with $\alpha = 1$ is \$2.58 with the *SPL* and \$1.99 when using the 10th percentile in every survey. The typical *SPL* is \$4.42 while a typical *q10* threshold is \$3.34. As expected, floors using $\alpha = 4$ are smaller than floors using $\alpha = 1$ as the former gives more weight to smaller values of consumption while the latter gives equal weight across all observations below the poverty line. Other thresholds and floors using income surveys are reported in Table D1.

It is plausible that smaller values of consumption are more important to estimate the floor of consumption. This is why the Ravallion formulation gives no weight to consumption values above the poverty line. In a similar vein, floors estimated on poorer countries might be a more relevant estimate of a universal floor. To that end, Table 3 also reports floors using recent (post 2000) surveys only from low-income countries (LICs). For this subsample, the minimum value of the permanent consumption floor ranges between \$0.10 with $z = q10$ and $\alpha = 4$, and \$0.63 when $z = IPL$ or *SPL* and $\alpha = 1$.

Table 3: Permanent consumption floor using various poverty lines and curvature

z	α	All surveys (N = 809)			Post-2000 surveys (N = 698)			LIC surveys (N = 164)		
		Min	p50	Mean	Min	p50	Mean	Min	p50	Mean
IPL	1	0.48	1.41	1.40	0.63	1.44	1.43	0.63	1.25	1.21
IPL	4	0.24	1.03	1.06	0.38	1.06	1.10	0.41	0.85	0.86
NPL	1	0.45	2.42	2.69	0.45	2.61	2.82	0.45	1.24	1.39
NPL	4	0.31	1.73	1.94	0.31	1.86	2.03	0.31	0.90	1.00
SPL	1	0.48	2.58	2.95	0.63	2.86	3.10	0.63	1.40	1.43
SPL	4	0.24	1.86	2.11	0.42	2.04	2.23	0.42	0.99	1.02
q10	1	0.04	1.99	2.36	0.16	2.27	2.52	0.16	0.84	0.88
q10	4	0.02	1.42	1.69	0.10	1.60	1.82	0.10	0.58	0.62
q25	1	0.13	2.56	3.08	0.25	2.96	3.28	0.25	1.12	1.15
q25	4	0.06	1.88	2.20	0.17	2.10	2.35	0.17	0.76	0.82

Note: This table reports the minimum, median, and mean value of consumption floors across surveys using the method proposed by Ravallion (2016) (see equation 1). The floors depend on a poverty line z and a curvature parameter α used for weighting. A linear weight is assumed when $\alpha = 1$, a quadratic weight with higher weights to the bottom values when $\alpha = 2$, and so on. The summary statistics from the distribution of floors in Figure 2 are reported in the first 3 columns of the first row ($z = \text{IPL}$ and $\alpha = 1$). Floors are expressed in daily per capita 2017 PPP\$. For floors using other poverty lines, curvature parameters and income surveys, see also Table D1.

Acronyms: z – poverty line; Min – minimum; p50 – median; IPL – International poverty line; NPL – national poverty line; SPL – societal poverty line; q10 – threshold income of the 10th percentile; q25 – threshold income of the 25th percentile.

4.2. Consumption floor using various parametric distributions

4.2.1. Consumption floor using the Beta distribution

The appeal for using the Beta distribution is the known functional form, which requires only the threshold parameter to be specified. However, unlike the floors estimated using Ravallion's approach, floors defined using the Beta distribution are not bounded at the bottom. Since Ravallion's floor is a weighted average of the consumption values of the bottom of the distribution, they are bounded at the bottom at the observed minimum value in the survey. The observed minimum is the extreme value of the floor when there is only one observation below the poverty threshold, or all the weight is assigned to this observation. This is not the case for Beta distribution. Indeed, depending on the poverty line used, between 1 percent and 10 percent of floors estimated with the Beta distribution approach are negative. Table 4 presents summary statistics of the distribution of consumption floors excluding non-positive values. The typical consumption floor ranges from \$0.69 to \$1.12 when using all consumption surveys, from \$0.71 to \$1.29 when using more recent (post-2000) surveys, and from \$0.31 to \$0.76 when restricting the sample to recent LIC surveys. The median value of the consumption floor is robust to the inclusion of negative values (see Table D2).

Table 4: Consumption floors using the Beta distribution

z	All surveys				Post-2000 surveys				LIC surveys			
	N	Min	p50	Mean	N	Min	p50	Mean	N	Min	p50	Mean
IPL	693	0.02	0.69	2.13	594	0.02	0.71	1.93	159	0.05	0.62	2.38
NPL	783	0.01	1.09	1.22	673	0.01	1.18	1.27	163	0.14	0.65	0.75
SPL	795	0.01	1.10	1.27	686	0.01	1.19	1.33	161	0.09	0.76	0.77
q10	721	<0.01	0.73	0.95	633	<0.01	0.80	1.00	142	<0.01	0.31	0.36
q25	797	<0.01	1.12	1.38	689	<0.01	1.29	1.48	161	<0.01	0.47	0.52

Note: This table reports the values of the consumption floor using the Beta distribution as outlined in equation 2. Negative floors are excluded. Floors are expressed in daily per capita 2017 PPP\$. For floors using other poverty lines and income surveys, as well as including negative floors, see also Table D2.

Acronyms: z - poverty line; N – number of surveys; Min – minimum; p50 – median; IPL – International poverty line; NPL – national poverty line; SPL – societal poverty line; q10 – threshold income of the 10th percentile; q25 – threshold income of the 25th percentile.

4.2.2. Floors using the three-parameter log normal and the Dagum III distributions

Cohen and Whitten (1980) outline various ways to estimate the three-parameter log normal. Among those they outline, they recommend using a version of the method of moments estimator (ME) when the third moment is less than one and a version of the localized maximum likelihood estimator (LMLE) when the third moment is greater or equal to one. Table 5 reports values of the floor using both approaches. Note that there are only three surveys that have skewness that is less than 1. About a third of the floors using the LMLE and a tenth of the floors using ME are non-positive. These floors are excluded from Table 5. Since TPLN and the Dagum III distributions have a lower bound that is strictly smaller than the smallest observation in the survey, for sensitivity, we also report the floors fitted by dropping observations below \$0.50 and \$1.00.

The median consumption floor using the LMLE ranges from \$0.50 for low-income countries to \$0.61 when considering all surveys or post-2000 surveys. The minimum value of the floor is much smaller, \$0.02 when considering all surveys and \$0.24 when considering just the recent LIC surveys. We were unable to fit the maximum likelihood estimator to all distributions, hence, the floors are only reported for 28 LIC surveys after 2000.¹¹ While we were able to apply the ME approach to a larger set of surveys, the results here must be considered in light of the recommendation from Cohen and Whitten (1980). The range of floors using the ME is generally higher than the floors using the LMLE. The median consumption floor using the ME ranges from \$0.96 for recent low-income countries to \$2.52 for all surveys after 2000. The minimum values of the floor are closer to zero.

¹¹ Non-convergence of maximum likelihood estimation is common in models with lower-bound parameters, particularly for the Dagum Type III and three-parameter lognormal distributions. The principal difficulty is that the support of the distribution depends on the parameter vector, creating a likelihood surface with hard boundaries and non-smooth regions. Poorly scaled data further aggravate the problem by yielding ill-conditioned Hessians and unstable numerical derivatives (Wang and Flournoy, 2015; Heathcote et al., 2024).

Table 5: Floors using the three-parameter log normal distribution

	N	Min	p50	Mean	N	Min	p50	Mean
	(a) LMLE, Full distribution				(b) LMLE, Drop < \$0.50			
All	349	0.02	0.61	0.69	349	0.02	0.61	0.69
Post-2000	327	0.02	0.61	0.70	327	0.02	0.61	0.70
LIC	28	0.24	0.50	0.53	28	0.24	0.50	0.53
	(c) LMLE, Drop < \$1.00				(d) ME			
All	372	0.02	0.76	0.75	747	<0.01	2.29	2.83
Post-2000	348	0.02	0.76	0.75	644	<0.01	2.52	3.01
LIC	28	0.45	0.80	0.79	150	<0.01	0.96	1.13

Note: This table reports values of the floor for the three-parameter log normal (TPLN) distribution using the local maximum likelihood estimator (LMLE) and the method of moments estimates (ME). Negative floors are excluded. Floors are expressed in daily per capita 2017 PPP\$. For details, see equation 3 and Appendix C. See Table D3 for income floors.

Acronyms: LIC – low-income country; N – number of surveys; Min – minimum; p50 – median.

Like the TPLN, the maximum likelihood estimator of the Dagum III distribution did not converge for all surveys. Among those that did converge, a third of the floors were negative.¹² Since Dagum III should strictly have positive floors, Table 6 presents result of the Dagum III for only positive floors. The median value of the floors estimated on consumption distributions ranges from \$0.41 for recent LIC surveys to \$0.73 for all surveys after 2000. The minimum values of the positive floor are close to zero.

In both the TPLN and Dagum III cases, dropping observations below \$0.50 and \$1.00 makes only very small difference. As expected, the floors get larger on average when the distribution is shifted right. Compared to the \$0.24 minimum LIC floor from the TPLN LMLE estimate, dropping observations below \$0.50 produces a minimum floor of \$0.24 and dropping observations below \$1.00 produces a minimum floor of \$0.45. For the same sample in the Dagum III distribution, the minimum floor is between \$0.02 and \$0.86 compared to \$0.02 for the full distribution.

¹² In models with an estimated lower bound—such as the three-parameter lognormal, the shifted Dagum (Type II), and the Dagum Type III—theoretical considerations require the floor parameter to be non-negative. In finite samples, however, maximum likelihood estimation may yield a negative estimated floor. This outcome reflects weak or boundary-driven identification: as the floor approaches the sample minimum from below, the likelihood often remains flat or changes only marginally, leading the optimizer to select values at or beyond the boundary. The issue is exacerbated by data scale, since poorly scaled variables (e.g., income measured in very large units or observations close to zero) can result in ill-conditioned likelihoods and imprecise gradients. In the Dagum Type III model, where the floor is not a free parameter but is implicitly determined by other parameters, small numerical perturbations can further translate into implausible implied lower bounds. Consequently, a negative estimated floor should be interpreted as a numerical artifact rather than evidence of negative support, with the effective support of the fitted distribution remaining anchored by the observed data.

Table 6: Floors using the Dagum III distribution

	(a) Full Distribution				(b) Drop < \$0.50				(c) Drop < \$1.00			
	N	Min	p50	Mean	N	Min	p50	Mean	N	Min	p50	Mean
All	343	<0.01	0.70	0.91	358	0.02	0.66	0.94	332	0.08	1.00	1.20
Post-2000	312	0.02	0.73	0.95	323	0.02	0.70	0.96	304	0.08	1.00	1.20
LIC	33	0.02	0.41	0.40	43	0.02	0.50	0.50	30	0.86	0.99	0.98

Note: This table reports values of the floor for the Dagum III distribution (see Appendix C). Negative floors are excluded. Floors are expressed in daily per capita 2017 PPP\$. See Table D3 for income floors.

Acronyms: LIC – low-income country; N – number of surveys; Min – minimum; p50 – median.

4.3. Floors using the outlier detection method

Floors can be defined using an outlier detection method similar to the LIS approach defined in Section 3.3. Table 7 reports consumption and income floors defined at k times the *IQR* distance below the first quartile of the distribution (see equation 3). LIS uses $k = 3$. Table 7 reports values for k between 2 and 4. The minimum value of the floors using this approach is quite small, in all cases less than \$0.05. The median among consumption surveys ranges from \$0.17 (when $k = 4$) to \$0.86 (when $k = 2$) using all surveys, from \$0.20 (when $k = 4$) to \$0.96 (when $k = 2$) when considering surveys after 2000, and from \$0.06 (when $k = 4$) to \$0.34 (when $k = 2$) when focusing only on low-income countries after 2000.

Table 7: Floors using LIS-type outlier detection method

k	All surveys				Post-2000 surveys				LIC surveys			
	N	Min	p50	Mean	N	Min	p50	Mean	N	Min	p50	Mean
2	806	<0.01	0.86	1.10	698	0.05	0.96	1.18	164	0.05	0.34	0.38
3	806	<0.01	0.37	0.53	698	0.01	0.44	0.57	164	0.01	0.14	0.18
4	806	<0.01	0.17	0.26	698	<0.01	0.20	0.28	164	<0.01	0.06	0.08

Note: This table reports statistics of the floor using the expression: $y_f = \exp(Q_1 - k \times (Q_3 - Q_1))$, where Q_1 is the 1st quartile, Q_3 is the 3rd quartile, and k is the distance parameter. While the table reports the floors using $k \in 2,3,4$, LIS uses $k = 3$. Floors are expressed in daily per capita 2017 PPP\$. See Table D3 for income floors and Table D4 for floors defined using other outlier-detection methods.

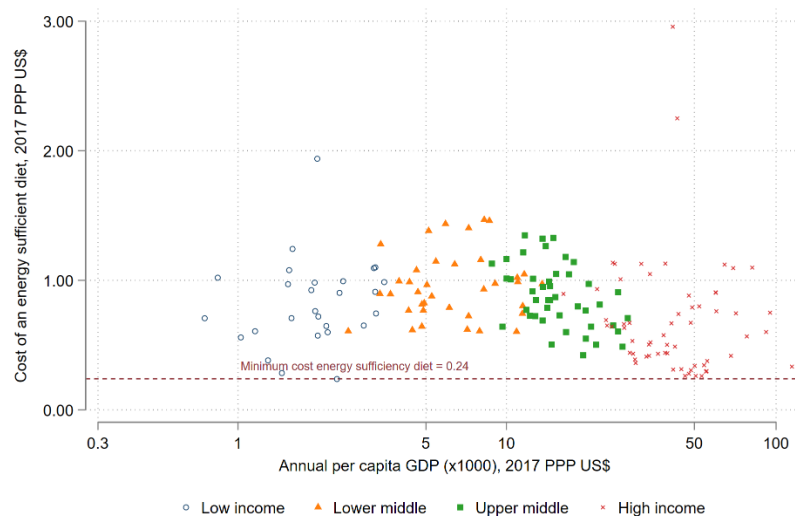
Acronyms: N – number of surveys; Min – minimum; p50 – median.

4.4. Minimum cost of energy sufficient diet

We complement the results based on statistical approaches with benchmarks obtained from the food poverty literature. Specifically, we look at the cross-country distribution of the cost of an energy sufficient diet as a monetary benchmark of short-term survival. Figure 3 reports the energy sufficient diet cost across countries in 2017. The costs range from \$0.24 per day to close

to \$3.00 per day. Surprisingly the cost of this diet is fairly stable across countries at various income levels with the average cost in the richest countries almost equal to the average cost in the poorest set of countries. The median cost among low-income countries is \$0.76 per day. If instead the focus was on the bottom 10 percent of the cheapest energy-sufficient diets (note this would include countries in the high-income category as well), the median floor is \$0.31 per day.

Figure 3: Distribution of the energy sufficient diet cost across countries by income level



Source: Food Prices for Nutrition 2.1, World Development Indicators (World Bank, n.d.) GDP per capita in 2017 PPP\$, World Development Indicators Archive (World Bank, n.d.),

Note: This figure reports the distribution of daily costs of an energy sufficient diet for various countries ranked by their GDP per capita. Costs and income are reported for the year 2017 and expressed in daily per capita 2017 PPP\$. The horizontal axis uses a logarithmic scale.

5. Summarizing the various floors

The bottom-coding threshold must **balance two trade-offs**. First, a floor that is too large implies that a large segment of the distribution is affected by the bottom code. Table D5 reports the share of the survey population affected by a \$0.25 and \$0.50 per day bottom code for the top 10 affected countries. For example, a bottom code of \$0.50 per day would mean altering up to 30 percent of the data in the Democratic Republic of Congo's 2004 survey, whereas a bottom code of \$0.25 per day would affect "only" 6 percent of the data. This would suggest that a lower bottom code is preferable.

A second trade-off to consider is that a floor that is too small may be mostly reflecting measurement errors. Additionally, very low values, constituting only a small part of the distribution, can become very influential for (bottom-sensitive) measures.

5.1. Evidence from the cost of energy sufficient diet and statistical approaches

The minimum cost of a diet consisting of the least expensive staple starches that results in 2,330 kcal per day across all countries is \$0.24 per person per day in 2017 PPP\$. While someone can survive on fewer calories for some periods of time, it should also be noted that this diet is not healthy as it only covers calories without accounting for other nutrients. Interviews conducted on a handful of families over a week in December 2020 in the Republic of Yemen revealed that people lived on 1,332 kcal per day during the conflict (Liskey 2021). This is roughly half of the recommended daily intake of 2,330 kcal. In the median low-income country, it would cost around \$0.38/day to consume 1,332 kcal per day (the median cost among LICs for 2,330 kcal is \$0.76/day).

With statistical approaches, the distribution of floors varies depending on the choice of method or underlying parameter. Therefore, it is essential to clearly define parameters. As lower consumption values are more prevalent in low-income countries, establishing a bottom-code based on low-income countries ensures this threshold remains appropriate for wealthier countries as well. For the Ravallion and Beta distribution methods, adopting the 10th percentile as the poverty line—as recommended by Beduk, Jara, and Ravallion (forthcoming)—seems reasonable. Table 8 reports estimate from post-2000 survey in low-income countries in column 3. The lowest value of the permanent consumption floor among low-income countries is \$0.16 with $\alpha = 1$. Likewise, the typical Beta distribution floor in low-income countries is \$0.31. The typical consumption floor for the same group is \$0.50 when considering the three-parameter log normal, is \$0.41 when considering the Dagum III distribution, and is \$0.14 using the LIS-type floors.

Table 8: Summary of the floors using various statistical approaches

Method	Value of consumption floor in daily 2017 PPPs					
	N	All surveys	N	Post-2000 surveys	N	LIC surveys
(a) Ravallion ($\alpha = 1$)	809	0.04	698	0.16	164	0.16
Ravallion ($\alpha = 4$)	809	0.02	698	0.10	164	0.10
(b) Beta	721	0.73	633	0.80	142	0.31
(c) TPLN, LMLE	350	0.61	327	0.61	28	0.50
(d) Dagum III	344	0.69	312	0.73	33	0.41
(e) LIS (3 x IQR below Q1)	809	0.36	698	0.44	164	0.14

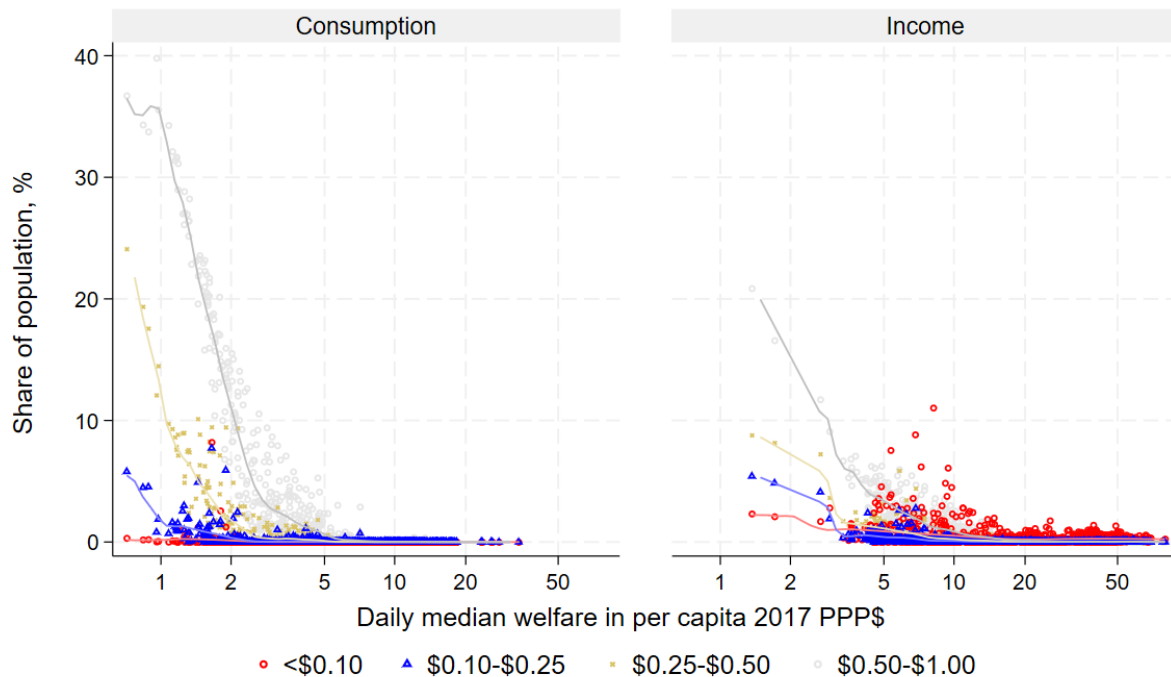
Note: This table summarizes consumption floors across various methods outlined in Section 4. Ravallion floor is the minimum value, while all other floors are median values. Ravallion and Beta floors are expressed for $z = q_{10}$ See Tables 3, 4, 5, 6, and 7. Floors are expressed in daily per capita 2017 PPP\$.

Acronyms: z - poverty line; q_{10} - threshold income of the 10th percentile; N - sample size.

5.2. Evidence from inspecting the data

One could also inspect the raw data for measurement concerns. One crude approach is to correlate the share of population with small values of income or consumption to the median welfare in the survey. Barring measurement concerns, we should expect the share of people with small values of consumption or income to decrease as median welfare rises. To see this, Figure 4, plots the distribution of consumption or income in each survey for those below \$0.10, between \$0.10 and \$0.25, between \$0.25 and \$0.50, and between \$0.50 and \$1.00 per day. The share of the population with those amounts of consumption or income is compared to the median of the survey. For the first two groups, there are some noticeable jumps. For instance, the share of people earning less than \$0.10 increases in many surveys when the daily median income is between \$5 and \$10. Compared to this, the share reporting below \$0.10 is much less in surveys with median income less than \$5. Similar erratic jumps are evident for the next higher bin. The share of people consuming more than \$0.10 but less than \$0.25 increases when the median consumption is around \$2; this proportion is much lower in surveys with median consumption around \$1. These patterns are not as visible once one moves to higher consumption and income bins, i.e., those above \$0.25 per day. The erratic jumps for the bins with lower welfare value could be a potential sign of greater measurement issues.

Figure 4: Share of population in income groups below various thresholds



Note: This figure reports the share of people living with various amounts of consumption or income in each survey expressed in daily 2017 PPP\$.

5.3. Evidence from contribution function

Another approach is to check the contribution of some parts of the distribution to an indicator. Since the coding is at the bottom of the distribution, we can take a distribution sensitive indicator and check how much the bottom x percent of the distribution contributes to the value of the indicator with and without various bottom codes. We then compare this contribution to the contribution on the same indicator using the x percent of a well-known distribution.

In what follows, we estimate the contribution of the bottom decile to the Prosperity Gap (Kraay et al. 2023) for a well-behaved distribution (the lognormal) and compare this with the contribution of the bottom decile across our sample with various bottom-coding thresholds.¹³

We can define the contribution of the p^{th} percentile to the Prosperity Gap as $S(p) \equiv \frac{\sum_{i=1}^p y_i^{-1}}{\sum_{i=1}^N y_i^{-1}}$ with income, y , and $i = 1, 2, \dots, N$ observations. For a lognormal distribution, $S(p) = \Phi(\sigma - \Phi^{-1}(1 - p))$, where p is the percentile of interest and standard deviation, $\sigma = \sqrt{2}\Phi^{-1}\left(\frac{1+G}{2}\right)$, and G is the Gini coefficient. This implies that the contribution of the bottom decile to the Prosperity Gap in a country with very high Gini coefficient (Gini index equal to 60) is 46 percent. Likewise, the bottom decile's contribution for a country with Gini of 40 is 30 percent. In our sample of 1,826 surveys, about 1 percent of surveys have a Gini index as high or higher than 60, 35 percent have a Gini between 40 and 60, and 65 percent have Gini less than 40. So, given a lognormal assumption, we should find a similar distribution of the bottom decile contribution to the calculation of the Prosperity Gap: in about 1 percent of cases the contribution would be more than 46 percent, in two thirds of cases would be less than 30 percent, and in about a third of the cases it would be between 30 percent and 46 percent.

Table 9 reports the contribution of the bottom decile to the Prosperity Gap for consumption and income surveys. The contributions are grouped into the share of surveys with less than 30 percent, between 30 percent and 45 percent, and higher than 45 percent contribution. In the raw data (panel a), we find that in 88 percent (26 percent) of consumption (income) surveys, the bottom decile contributes less than 30 percent to the Prosperity Gap. We also see that in 1 percent of consumption surveys and 22 percent of income surveys, the bottom decile contributes more than 45 percent to the Prosperity Gap.

Bottom coding at any level limits the number of surveys with contributions in excess of 45 percent. For instance, bottom coding at \$0.10 brings down the share of income surveys with contributions over 45 percent from 22 percent to 13 percent. If income surveys more closely follow a lognormal distribution, then it could be argued that the higher the bottom code, the more closely it mimics the distribution of Ginis in our sample. For instance, a bottom code of

¹³ We use the Prosperity Gap, which is equivalent to the inverse of the Atkinson(2) and shares the same distribution sensitive properties as outlined in the introduction.

\$0.50 would imply that 35 percent of income surveys have contributions less than 30 percent to the Prosperity Gap, 62 percent of income surveys have contributions between 30 percent and 40 percent, and only 3 percent of income surveys have contributions above 45 percent.

Table 9: The distribution of contribution of the bottom decile to the Prosperity Gap

	Share of consumption surveys			Share of income surveys		
	below 30%	30% - 45%	above 45%	below 30%	30% - 45%	above 45%
(a) Raw data	88.4	10.5	1.1	25.8	52.5	21.8
(b) Threshold						
\$0.10	88.6	10.8	0.6	27.8	59.0	13.3
\$0.15	89.1	10.5	0.4	28.6	61.0	10.3
\$0.20	89.7	10.0	0.2	29.5	62.1	8.4
\$0.25	90.4	9.5	0.1	30.3	62.4	7.3
\$0.30	91.3	8.7	0.0	31.0	62.8	6.2
\$0.35	92.1	7.9	0.0	31.9	62.9	5.2
\$0.40	93.0	7.0	0.0	32.6	63.0	4.4
\$0.45	93.7	6.3	0.0	33.1	62.8	4.1
\$0.50	94.2	5.8	0.0	34.5	62.2	3.2

Note: This table reports the share of surveys with the bottom decile contributing less than 30%, between 30% and 45%, and over 45% to the Prosperity Gap in the survey.

5.4. Summary and proposing \$0.25 per day

Triangulating various evidence and with the constraints (too high or too low bottom code) in mind, a threshold of \$0.25 per day seems reasonable to use as a bottom code from the range of methods discussed above. A threshold above \$0.50 per day would be too intrusive on the surveys considering that the higher end of permanent consumption ranges around this threshold. The range of values between \$0.25 and \$0.50 limits observable measurement errors, while a threshold below \$0.25 per day does not sufficiently limit the contribution of the bottom on distribution sensitive indicators compared to a well-behaved distribution. The \$0.25 per day threshold is also almost equivalent to the minimum of the least-cost energy sufficient diet, and it is within the range of the typical floors (ranging between \$0.14 and \$0.50) estimated among recent LIC surveys using various statistical approaches.

Given the wide range of estimates of the floor with various methodologies, a bottom code defined on each survey could be preferable. However, even for the same country, the range of estimated floors can be quite different depending on the survey if one were to use such an approach. More importantly, bottom codes that vary across countries or within a country across time imply a range of possible outcomes for most poverty and inequality measures. For example, the

Prosperity Gap estimated using the \$25 a day prosperity standard and a bottom code at \$0.25 per day has a maximum factor of 100, while using a bottom code at \$0.50 per day with the same prosperity standard will have a maximum factor of 200. Likewise, the Watts Index has a domain between 0 and 2.15 or 0 and 1.46 depending on the choice of \$0.25 or \$0.50 to bottom code with a poverty line at \$2.15. Comparing estimates defined on indicators that have different domains is not ideal.

6. Impact of the \$0.25 per day bottom code

6.1. Share of population below various bottom codes and the differences in welfare measures

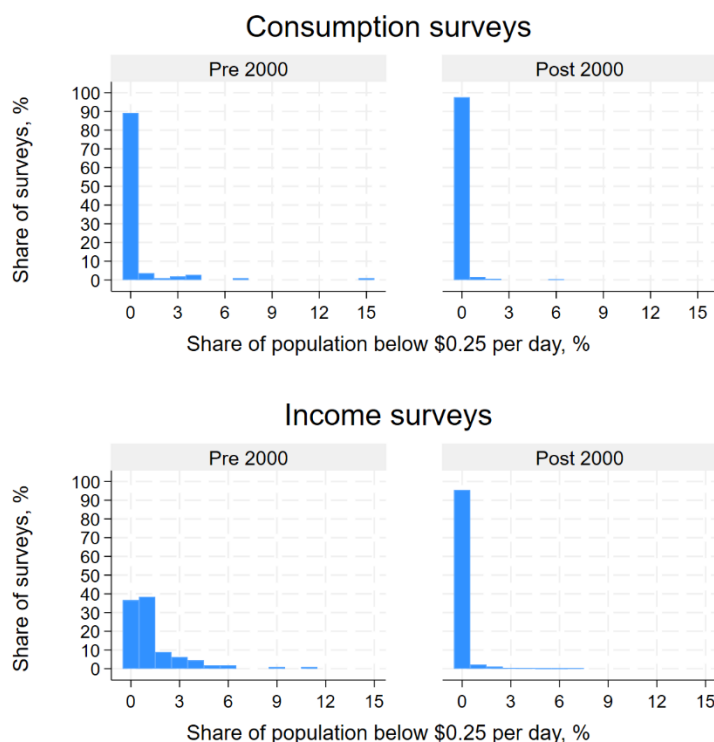
Figure 5 shows the share of population below the \$0.25 per person per day in consumption and income surveys separated into pre- and post-2000 surveys. While there are extreme cases, for example, the Zambia 1991 survey has over 15 percent of the population below \$0.25 per day, this is an exception rather than a norm. Close to two-thirds of all surveys have no observations below \$0.25 per day. Less than 1 percent of the population lives below \$0.25 in over 95 percent of post-2000 surveys and at least 90 percent of all surveys in the GMD. While more than half of the pre-2000 income surveys have some part of the distribution below \$0.25, only 14 surveys or 11 percent of pre-2000 income surveys have more than 3 percent of the population below the bottom code. Across all years, the maximum share of the population below \$0.25 is 16 percent in Zambia's 1991 consumption survey and 11 percent in Suriname's 1999 income survey.

6.2. Sensitivity of poverty and inequality indicators to various thresholds

Table 10 reports the difference in mean income, FGT(1), FGT(2), and the Watts index due to the bottom code for the ten surveys with the most observations below \$0.25 per day. Panel a reports the estimates for consumption surveys and panel b reports the same for income surveys. The difference in estimated values of the mean income is less than 0.5 percent in most cases with the highest difference of 0.7 percent in Zambia's 1991 survey where the mean income before the bottom code is \$3.10 and after the bottom code is \$3.12. Zambia's 1991 survey is also the survey with the largest share of the population living below \$0.25 – close to 16 percent. Likewise, the difference in the FGT(1) and FGT(2) measures are largest for the Zambia 1991 survey with a 2.9 percent decline in FGT(1) with the bottom code and a 6.9 percent decline in FGT(2) with the bottom code. Watts index is the most sensitive to the changes in the bottom tail of the measures reported in Table 10. The largest decline in the Watts index is for Zambia in the 1991 survey where the index decreases from 93.3 to 72.3 (22.5 percent). The changes in the income survey are relatively smaller than the changes in the consumption surveys. While Suriname's 1999 survey has the largest share of the population earning below \$0.25 – 11 percent – it has the largest

decline only in the Watts index of 17.8 percent with the bottom code. Micronesia in 1991 has the largest decline in FGT(1) and FGT(2) with 1.2 percent and 3.1 percent, respectively.

Figure 5: Distribution of share-of-population-below \$0.25 per day across surveys



Note: This figure reports the share of population in 809 consumption and 1017 income surveys with observations below \$0.25 per day.

For the same set of countries, Table 11 reports the difference in inequality measures including the Gini index, mean log deviation (MLD), Atkinson inequality index with $\varepsilon = 2$ (AI(2)), and the Prosperity Gap due to the bottom code at \$0.25 per day. The differences in the Gini index are the smallest and mainly limited to consumption surveys. Again, the largest difference is for Zambia in 1991 where the Gini prior to the bottom code is 60.5 and after is 59.4 (1.8 percent decline). As expected, the differences for the other bottom-sensitive measures are larger. For example, the Prosperity Gap for the Zambia 1991 survey changes from 212 before the bottom code to 33 after (84 percent lower), the AI(2) changes from 0.962 to 0.757 (21 percent lower) and the MLD changes from 0.937 before to 0.735 after (22 percent lower). It is important to note, however, that the rank correlation between the Prosperity Gap, a measure that can be decomposed into the mean and inequality, and the mean income is much closer with the bottom code compared to without, implying that the rankings are more in line with expectations and not influenced by a few very small observations.

Table 10: Various poverty and welfare measures with and without bottom code at \$0.25 per day

Country	Year	Pop below \$0.25, %	Mean	Mean*	Diff, %	P1	P1*	Diff, %	P2	P2*	Diff, %	Watts	Watts*	Diff, %
(a) Consumption surveys														
Zambia	1991	15.9	3.1	3.1	0.7%	37.0	36.0	-2.9%	28.7	26.7	-6.9%	93.3	72.3	-22.5%
Lesotho	1994	7.1	4.3	4.3	0.1%	31.1	30.9	-0.9%	21.9	21.4	-2.3%	61.1	57.5	-5.9%
Lesotho	2002	6.1	2.3	2.3	0.2%	35.9	35.6	-0.7%	24.0	23.5	-2.0%	66.9	63.5	-5.0%
Congo, Dem. Rep.	2004	6.1	1.0	1.0	0.4%	58.3	58.3	-0.3%	41.5	41.1	-0.8%	109.6	107.6	-1.8%
Mozambique	1996	4.8	1.6	1.6	0.2%	49.9	49.8	-0.3%	34.4	34.2	-0.7%	91.3	89.8	-1.6%
Mali	1994	4.7	1.4	1.4	0.2%	51.4	51.2	-0.2%	35.5	35.3	-0.6%	93.9	92.4	-1.5%
Uzbekistan	1998	4.3	2.4	2.5	0.1%	27.7	27.6	-0.5%	17.2	17.0	-1.4%	44.8	42.5	-5.1%
Gambia, The	1998	3.6	1.9	1.9	0.2%	39.1	38.9	-0.4%	24.8	24.5	-1.2%	68.0	65.8	-3.2%
Burundi	1998	3.4	1.7	1.7	0.2%	39.9	39.8	-0.4%	24.7	24.4	-1.0%	68.2	66.3	-2.7%
Benin	2015	3.0	3.0	3.0	0.1%	23.0	22.9	-0.5%	14.6	14.4	-1.5%	41.7	40.2	-3.7%
(b) Income surveys														
Suriname	1999	11.0	13.1	13.1	0.0%	13.9	13.9	-0.3%	12.6	12.5	-0.6%	6.3	5.2	-17.8%
Panama	1989	9.8	11.9	11.9	0.0%	15.6	15.5	-0.3%	13.2	13.1	-0.7%	12.8	12.2	-4.8%
Colombia	2000	8.0	10.0	10.0	0.0%	11.3	11.3	-0.2%	9.6	9.6	-0.4%	6.4	6.1	-4.4%
Haiti	2001	7.7	2.7	2.7	0.3%	37.5	37.1	-1.1%	25.4	24.6	-2.9%	72.2	66.6	-7.7%
Panama	1991	7.0	12.7	12.7	0.0%	13.2	13.2	-0.3%	10.7	10.6	-0.7%	13.0	12.4	-4.1%
Haiti	2012	7.0	3.5	3.5	0.2%	31.7	31.4	-0.9%	21.6	21.0	-2.6%	60.4	55.8	-7.6%
Panama	1995	6.7	16.2	16.2	0.0%	10.5	10.5	-0.2%	8.9	8.8	-0.5%	8.2	7.9	-4.1%
Micronesia	2000	5.8	5.6	5.6	0.1%	24.5	24.2	-1.2%	17.1	16.6	-3.1%	49.0	44.8	-8.5%
Panama	1998	5.3	16.1	16.1	0.0%	9.4	9.4	-0.3%	7.7	7.6	-0.6%	8.3	8.0	-4.0%
Colombia	1999	5.2	9.0	9.0	0.0%	11.5	11.5	-0.2%	8.6	8.6	-0.5%	11.7	11.4	-2.4%

Note: This table reports differences (in percent) in the mean, FGT1 (P1), FGT2 (P2), and Watts index (Watts) for indicators calculated without a bottom code and those calculated with a bottom code at \$0.25 (*). Negative observations are excluded throughout. Panel (a) reports the top 10 consumption surveys affected by the bottom code and panel (b) reports the same for income surveys. *Acronym*: Diff – difference.

Table 11: Various inequality estimates with and without bottom code at \$0.25 per day

Country	Year	Pop below \$0.25, %	Gini	Gini*	Diff, %	MLD	MLD*	Diff, %	AI(2)	AI(2)*	Diff, %	PG	PG*	Diff, %
(a) Consumption surveys														
Zambia	1991	15.9	60.5	59.4	-1.8%	0.937	0.735	-21.6%	0.962	0.757	-21.3%	212.0	33.0	-84.5%
Lesotho	1994	7.1	63.2	63.0	-0.3%	0.839	0.804	-4.1%	0.882	0.774	-12.3%	49.1	25.5	-48.0%
Lesotho	2002	6.1	51.6	51.2	-0.7%	0.511	0.480	-6.1%	0.669	0.600	-10.3%	33.3	27.5	-17.4%
Congo, Dem. Rep.	2004	6.1	42.2	41.6	-1.2%	0.301	0.285	-5.4%	0.435	0.398	-8.4%	44.1	41.2	-6.4%
Mozambique	1996	4.8	53.6	53.3	-0.5%	0.498	0.485	-2.6%	0.574	0.551	-4.0%	37.0	35.0	-5.3%
Mali	1994	4.7	50.4	50.2	-0.6%	0.437	0.424	-2.8%	0.535	0.509	-4.8%	37.8	35.7	-5.4%
Uzbekistan	1998	4.3	43.7	43.6	-0.4%	0.349	0.327	-6.3%	0.662	0.507	-23.4%	29.7	20.3	-31.5%
Gambia, The	1998	3.6	48.5	48.3	-0.6%	0.422	0.402	-4.8%	0.579	0.517	-10.7%	30.7	26.8	-13.0%
Burundi	1998	3.4	42.3	42.1	-0.6%	0.326	0.309	-5.1%	0.514	0.444	-13.6%	31.1	27.1	-12.7%
Benin	2015	3.0	47.8	47.6	-0.3%	0.446	0.431	-3.3%	0.638	0.591	-7.3%	23.0	20.4	-11.5%
(b) Income surveys														
Suriname	1999	11.0	52.8	52.8	0.0%	0.430	0.418	-2.9%	0.882	0.682	-22.6%	14.5	5.37	-62.9%
Panama	1989	9.8	55.0	55.0	0.0%	0.540	0.533	-1.2%	0.799	0.768	-3.8%	9.53	8.27	-13.2%
Colombia	2000	8.0	55.3	55.3	0.0%	0.481	0.478	-0.6%	0.680	0.655	-3.8%	7.23	6.69	-7.5%
Haiti	2001	7.7	59.5	59.0	-0.8%	0.692	0.640	-7.6%	0.774	0.673	-13.1%	41.5	28.6	-31.2%
Panama	1991	7.0	55.5	55.5	0.0%	0.581	0.576	-1.0%	0.795	0.771	-3.0%	9.03	8.09	-10.5%
Haiti	2012	7.0	60.6	60.3	-0.5%	0.732	0.688	-6.0%	0.820	0.714	-12.8%	39.6	24.9	-37.0%
Panama	1995	6.7	55.1	55.1	0.0%	0.553	0.550	-0.6%	0.783	0.761	-2.9%	6.68	6.05	-9.4%
Micronesia	2000	5.8	63.3	63.1	-0.3%	0.857	0.817	-4.7%	0.857	0.788	-8.0%	31.3	21.1	-32.5%
Panama	1998	5.3	55.4	55.4	0.0%	0.566	0.562	-0.6%	0.776	0.753	-2.9%	6.59	5.99	-9.1%
Colombia	1999	5.2	56.8	56.8	0.0%	0.561	0.558	-0.5%	0.708	0.695	-1.9%	9.11	8.71	-4.4%

Note: This table reports differences (in percent) in the Gini index, mean log deviation (MLD), Atkinson inequality index with $\varepsilon = 2$ (AI(2)), and Prosperity Gap (PG) for indicators calculated without a bottom code and those calculated with a bottom code at \$0.25 (*). Negative observations are excluded throughout. Panel (a) reports the top 10 consumption surveys affected by the bottom code and panel (b) reports the same for income surveys. *Acronym*: Diff – difference.

7. Conclusion

Distribution-sensitive poverty and welfare measures are highly responsive to low individual consumption or income values. However, this sensitivity can lead to excessive influence from very small, potentially erroneous values, thereby complicating welfare comparisons. For instance, the Atkinson index is notably impacted by even a small percentage of low-consumption observations. Measurement errors further exacerbate these concerns. Organizations address such data issues by bottom-coding or truncating distributions, though there is no consensus on the appropriate threshold.

This paper, utilizing more than 800 consumption surveys, systematically evaluates potential thresholds by triangulating statistical methods and biological consumption minimums. The least cost of a bare-minimum diet of staple starches providing 2,330 kcal daily cost \$0.24 per person across countries in 2017. Consumption floors for low-income countries with a poverty threshold at the 10th percentile of the distribution range from \$0.14 (using LIS-type approach) to \$0.50 (Three Parameter Log Normal), with the minimum of Ravallion-type floor at \$0.16 and the Beta distribution floor at \$0.32.

Based on this, we propose a bottom-coding threshold of \$0.25 per person per day for consumption. This threshold only meaningfully impacts a handful of surveys, most of which are more than two decades old.

This paper centers on identifying an appropriate bottom-coding threshold for consumption distributions, acknowledging that bottom-coding income presents unique challenges since measured income can be zero or negative while consumption must be larger than zero. Negative or extremely low income values complicate the calculation of poverty and welfare measures and introduce measurement errors, as households with negative incomes are potentially relatively well-off. Consequently, substituting these values with a low bottom-coding threshold can underestimate both the well-being of such households and overall inequality. The broader issues concerning a bottom-code for income are not addressed here. Instead, this paper focuses on determining an appropriate bottom-coding threshold for consumption distributions. While recognizing the distinct challenges associated with bottom-coding income—given the potential for zero or negative measured values—we demonstrate the effect of a \$0.25 per person per day threshold on non-negative incomes, emphasizing the limitations of this approach and reserving the broader discussion of income bottom-coding for future research.

References

- Aristizabal, R. J. (2012). Estimating the parameters of the three-parameter lognormal distribution.
- Bedük, Selçuk, H. Xavier Jara, Martin Ravallion. (forthcoming) "No one left behind: how consistent are cross-sectional and longitudinal estimates of the income floor?"
- Belotti, F., Mancini, G., & Vecchi, G. (2024). Outlier detection for inequality and poverty analysis. *The Stata Journal*, 24(4), 630-665.
- Cohen Jr, A. C. (1951). Estimating parameters of logarithmic-normal distributions by maximum likelihood. *Journal of the American statistical association*, 46(254), 206-212.
- Cohen, A. C., & Whitten, B. J. (1980). Estimation in the three-parameter lognormal distribution. *Journal of the American Statistical Association*, 75(370), 399-404.
- Cowell, F. A., & Flachaire, E. (2007). Income distribution and inequality measurement: The problem of extreme values. *Journal of Econometrics*, 141(2), 1044-1072.
- Cowell, F. A., & Victoria-Feser, M. P. (1996). Robustness properties of inequality measures. *Econometrica: journal of the Econometric Society*, 77-101.
- Dagum, C. (1980). The generation and distribution of income, the Lorenz curve and the Gini ratio. *Economie Appliquee*, 33, 327-367.
- FAO, IFAD, UNICEF, WFP and WHO. (2020). The State of Food Security and Nutrition in the World 2020: Transforming food systems for affordable healthy diets. Rome, FAO. <https://doi.org/10.4060/ca9692en>
- Foster, E., Jolliffe, D. M., Ibarra, G. L., Lakner, C., & Baah, S. K. T. (2025). Global poverty revisited using 2021 PPPs and new data on consumption. *Policy Research Working Paper Series*, 11137.
- Heathcote, A., Brown, S., & Cousineau, D. (2004). QMPE: Estimating Lognormal, Wald, and Weibull RT distributions with a parameter-dependent lower bound. *Behavior Research Methods, Instruments, & Computers*, 36, 277–290. <https://doi.org/10.3758/BF03195574>
- Herforth, A., Venkat, A., Bai, Y., Costlow, L., Holleman, C., & Masters, W. A. (2022). Methods and options to monitor the cost and affordability of a healthy diet globally Background paper for The State of Food Security and Nutrition in the World 2022.
- Hlasny, V., Ceriani, L., & Verme, P. (2022). Bottom incomes and the measurement of poverty and inequality. *Review of Income and Wealth*, 68(4), 970-1006.
- Jenkins, S. P., Burkhauser, R. V., Feng, S., & Larrimore, J. (2011). Measuring inequality using censored data: a multiple-imputation approach to estimation and inference. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 174(1), 63-81.
- Jenkins, S. P., & Jantti, M. (2005). Methods for summarizing and comparing wealth distributions. *Working Paper 2005-05*. ISER, 2005.

- Jolliffe^(r), D., Mahler^(r), D. G., Lakner^(r), C., Atamanov^(r), A., & Tetteh-Baah, S. K. (2025). Poverty and prices: Assessing the impact of the 2017 PPPs on the international poverty line and global poverty. *The World Bank Economic Review*, 39(3), 497-521.
- Jolliffe, D., Margitic, J., Ravallion, M., & Tiehen, L. (2024). Food stamps and America's poorest. *American Journal of Agricultural Economics*, 106(4), 1380-1409.
- Jolliffe, D., & Prydz, E. B. (2021). Societal poverty: A relative and relevant measure. *The World Bank Economic Review*, 35(1), 180-206.
- Kleiber, C. (2007). On the Zenga order within parametric families of income distributions. Working paper, Universitat Basel, Switzerland.
- Kleiber, C. (2008). A guide to the Dagum distributions. In *Modeling income distributions and Lorenz curves* (pp. 97-117). New York, NY: Springer New York.
- Kraay, A., Lakner, C., Ozler, B., Decerf, B., Jolliffe, D., Sterck, O., & Yonzan, N. (2024). A simple decomposable distribution sensitive welfare index. *World Bank Working Paper*.
- Liskey, Amber (2021). Bread and Tea: A look into Yemen's Food Insecurity Crisis. https://egyptssp.ifpri.info/2021/07/01/bread-and-tea-a-look-into-yemens-food-security-crisis/#_edn1
- Meyer, B.D., & Sullivan, J.X. (2003). Measuring the well-being of the poor using income and consumption. *Journal of Human Resources*, 38, S1180–220.
- Neuenschwander, J. (2020). *Top and bottom coding at LIS*. LIS Technical Working Paper 9, September. <https://www.lisdatacenter.org/wps/techwps/9.pdf>.
- Ravallion, M. (2016). Are the world's poorest being left behind?. *Journal of Economic Growth*, 21(2), 139-164.
- Tufts University (n.d.) Food Prices for Nutrition: Diet cost metrics for a better fed world. Available at: <https://sites.tufts.edu/foodpricesfornutrition/> (Accessed: 9 January 2026)
- Wang, H., & Flourney, N. (2015). *On the consistency of the maximum likelihood estimator for the three-parameter lognormal distribution*. Retrieved from <https://ossifragus.github.io/pdfs/lognormal.pdf>
- Wicksell, S. D. (1917). *The Correlation Function of Type A, and the Regression of Its Characteristics* (Vol. 58). Almqvist & Wiksells boktryckeri-a.-b.
- World Bank. (2020). Purchasing Power Parities and the Size of World Economies: Results from the 2017 International Comparison Program. Washington, DC: World Bank. Doi: 10.1596/978-1-4648-1530-0. License: Creative Commons Attribution CC BY 3.0 IGO
- World Bank (2023). *Ending poverty on a livable planet: Report to governors on World Bank evolution*. Development Committee, World Bank Group.

<https://www.devcommittee.org/content/dam/sites/devcommittee/doc/documents/2023/Final%20Updated%20Evolution%20Paper%20DC2023-0003.pdf>

World Bank. (2024). Poverty and Inequality Platform (version 20240627_2017_01_02_PROD) [Data set]. World Bank Group. <https://pip.worldbank.org/>

World Bank. n.d. "World Bank Country and Lending Groups" World Bank. Accessed September 4th 2024. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>.

World Bank, n.d., World Development Indicators. Accessed February 12, 2024, <https://databank.worldbank.org/source/food-prices-for-nutrition#>

World Bank, n.d., World Development Indicators Archives. Accessed February 5, 2026. <https://databank.worldbank.org/source/wdi-database-archives>

Appendix A: Beta distribution floor

Assume that consumption between the consumption floor and poverty line z follow a shifted and rescaled Beta distribution, i.e., $y = y_f + (z - y_f)x$ with $x \sim \text{Beta}(\eta, \beta)$. As long as z is below the modal value of the overall density of the distribution, it is reasonable to impose the further restrictions that $\eta > 1$ and $\beta = 1$. Together, this ensures that the density of y is monotonically increasing over the interval $[y_f, z]$. The moment conditions for the Beta distribution with $\eta > 1$ and $\beta = 1$ are as follows:

$$E[y] = y_f + (z - y_f) \frac{\eta}{\eta + 1}$$

$$\text{var}[y] = (z - y_f)^2 \frac{\eta}{(\eta + 1)^2(\eta + 2)}.$$

The poverty gap, P_1 , is

$$P_1 = \frac{1}{N} \sum_i^q \left(1 - \frac{y_i}{z}\right)$$

where, N is the total number of individuals i , $Q \in 1, 2, \dots, q$ is the total number of poor, thus $P_0 = Q/N$, and z is poverty line. P_1 can be rewritten as

$$P_1 = \left(\frac{1}{N} \sum_i^q 1\right) - \left(\frac{1}{z} \times \frac{1}{N} \sum_i^q y_i\right)$$

$$P_1 = H - \left(\frac{1}{z} \times \frac{Q}{N} \times \frac{1}{Q} \sum_i^q y_i\right) = P_0 - \frac{P_0}{z} E[y]$$

$$P_1 = P_0 \left(1 - \frac{E[y|y < z]}{z}\right). \quad (i)$$

Denote that $E[y]$ as $E[y|y < z]$, which is the mean income of those who are poor ($y < z$) and P_0 is the headcount ratio of the poor. The ratio of poverty gap to headcount then is $\frac{P_1}{P_0} = \left(1 - \frac{E[y]}{z}\right)$. Using this ratio in equation (i), we can rewrite the expectation as $E[y] = z \left(1 - \frac{P_1}{P_0}\right)$.

Likewise, the squared poverty gap, P_2 , can be written as

$$P_2 = \left(\frac{1}{N} \sum_i^q \left(1 - \frac{y_i}{z}\right)^2\right) = \frac{1}{N} \sum_i^q 1 - \frac{2}{z} \frac{1}{N} \sum_i^q y_i + \left(\frac{1}{z^2} \times \frac{1}{N} \sum_i^q y_i^2\right)$$

$$P_2 = P_0 - \frac{2E[y]P_0}{z} + \frac{E[y^2]P_0}{z^2}.$$

Hence the ratio of P_2 to headcount can be rewritten as

$$\begin{aligned}\frac{P_2}{P_0} &= 1 - \frac{2E[y]}{z} + \frac{E[y^2]}{z^2} \\ \frac{P_2}{P_0} - 1 &= -\frac{2E[y]}{z} + \frac{E[y^2]}{z^2}\end{aligned}$$

and

$$E[y^2] = \frac{P_2 z^2}{P_0} - z^2 + 2E[y]z.$$

The variance then can be written as

$$\begin{aligned}var(y) &= E[y^2] - E[y]^2 = \frac{P_2 z^2}{P_0} - z^2 + 2E[y]z - z^2 + \frac{2z^2 P_1}{P_0} - \frac{z^2 P_1^2}{P_0^2} \\ var(y) &= \frac{P_2 z^2}{P_0} - z^2 + 2\left(z - \frac{z P_1}{P_0}\right)z - z^2 + \frac{2z^2 P_1}{P_0} - \frac{z^2 P_1^2}{P_0^2} \\ var(y) &= \frac{P_2 z^2}{P_0} - z^2 + 2z^2 - 2\frac{z^2 P_1}{P_0} - z^2 + \frac{2z^2 P_1}{P_0} - \frac{z^2 P_1^2}{P_0^2} \\ var(y) &= \frac{P_2 z^2}{P_0} - \frac{z^2 P_1^2}{P_0^2} = z^2 \left(\frac{P_2}{P_0} - \frac{P_1^2}{P_0^2} \right).\end{aligned}$$

Solving for the floor using the Beta distribution using the moment conditions,

$$\begin{aligned}E[y] &= y_f + \frac{(z - y_f)\eta}{\eta + 1} \\ z\left(1 - \frac{P_1}{P_0}\right) &= \frac{\eta y_f + y_f + \eta z - \eta y_f}{\eta + 1} \\ (P_0 z - P_1 z)(\eta + 1) &= (y_f + \eta z)P_0 \\ \eta P_0 z + P_0 z - \eta P_1 z - P_1 z &= y_f P_0 + \eta z P_0 \\ P_0 z - \eta P_1 z - P_1 z &= y_f P_0 \\ y_f &= z - \frac{\eta P_1 z}{P_0} - \frac{P_1 z}{P_0} = z\left(1 - (1 + \eta)\frac{P_1}{P_0}\right)\end{aligned}$$

$$\text{Or, } (z - y_f) = \frac{(\eta + 1)P_1 z}{P_0}$$

Using the variance from the Beta distribution,

$$var(y) = \frac{(z - y_f)^2 \eta}{(\eta + 1)^2 (\eta + 2)}$$

$$\frac{P_2 z^2}{P_0} - \frac{z^2 P_1^2}{P_0^2} = \frac{(\eta + 1)^2 P_1^2 z^2 \eta}{(\eta + 1)^2 (\eta + 2) P_0^2}$$

$$(P_0 P_2 z^2 - z^2 P_1^2)(\eta + 2) = P_1^2 z^2 \eta$$

$$\eta P_0 P_2 z^2 + 2 P_0 P_2 z^2 - \eta z^2 P_1^2 - 2 z^2 P_1^2 = \eta z^2 P_1^2$$

Which gives the value for alpha,

$$\eta = \frac{2 z^2 P_1^2 - 2 z^2 P_0 P_2}{P_0 z^2 P_2 - 2 z^2 P_1^2} = \frac{2 P_1^2 - 2 P_0 P_2}{P_0 P_2 - 2 P_1^2}$$

Solving for the floor,

$$y_f = z - \frac{\eta P_1 z}{P_0} - \frac{P_1 z}{P_0}$$

$$y_f = z - \left(\frac{2 P_1^2 - 2 P_0 P_2}{P_0 P_2 - 2 P_1^2} \right) \frac{P_1 z}{P_0} - \frac{P_1 z}{P_0}$$

$$= z - \left(1 + \frac{2 P_1^2 - 2 P_0 P_2}{P_0 P_2 - 2 P_1^2} \right) \frac{P_1 z}{P_0}$$

$$= z - \left(\frac{P_0 P_2 - 2 P_1^2 + 2 P_1^2 - 2 P_0 P_2}{P_0 P_2 - 2 P_1^2} \right) \frac{P_1 z}{P_0}$$

$$= z - \frac{- P_0 P_2}{P_0 P_2 - 2 P_1^2} \frac{P_1 z}{P_0}$$

$$y_f = z \left(1 + \frac{P_0 P_1 P_2}{P_0^2 P_2 - 2 P_0 P_1^2} \right).$$

Appendix B: Three-parameter log normal

Following Cohen & Whitten (1980) the likelihood function can be expressed as

$$L(x; \gamma, \mu, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \cdot \prod_1^n (x_i - \gamma)^{-1} \cdot \exp \left\{ -\frac{[\ln(x_i - \gamma) - \mu]^2}{2\sigma^2} \right\},$$

where, $\sigma > 0$, $-\infty < \mu < \infty$, and $\gamma < x < \infty$. μ is the mean, σ^2 is the variance, and γ is a threshold parameter. Then the function to maximize is

$$\ell(\gamma, \mu, \sigma) = - \sum_{i=1}^N \ln(x_i - \gamma) - N \ln \sigma^2 - \frac{N}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n [\ln(x_i - \gamma) - \mu]^2.$$

Appendix C: Dagum III distribution

Following Kleiber (2007), the probability density function can be expressed as

$$f(x) = \frac{apx^{ap-1}}{b^{ap} \left[1 + \left(\frac{x}{b} \right)^a \right]^{p+1}},$$

where, $a, b, p > 0$ and $0 < x_0 \leq x < \infty$. The maximum likelihood function of the shifted distribution can be expressed as follows:

$$\begin{aligned} \ell(a, b, p, x_0) = & N \ln a + N \ln p + aN \ln b - (a + 1) \sum_{i=1}^N \ln(x_i - x_0) \\ & - (p + 1) \sum_{i=1}^N \ln \left[1 + \left(\frac{b}{x_i - x_0} \right)^a \right], \quad x_i > x_0. \end{aligned}$$

Appendix D: Other results

Table D1: Permanent floors using Ravallion (2016)

z	α	All surveys				Post-2000 surveys				LIC surveys			
		N	Min	p50	Mean	N	Min	p50	Mean	N	Min	p50	Mean
(a) Consumption surveys													
q05	1	809	0.01	1.69	1.99	698	0.11	1.93	2.13	164	0.11	0.70	0.75
q05	4	809	0.01	1.20	1.44	698	0.07	1.34	1.55	164	0.07	0.47	0.52
q50	1	809	0.36	3.31	4.01	698	0.36	3.73	4.27	164	0.36	1.50	1.52
q50	4	809	0.19	2.44	2.88	698	0.24	2.74	3.07	164	0.24	1.03	1.09
(b) Income surveys													
IPL	1	591	0.00	0.92	0.91	479	0.00	0.95	0.95	3	0.69	0.70	0.82
IPL	4	591	0.00	0.40	0.44	479	0.00	0.43	0.47	3	0.38	0.39	0.47
NPL	1	729	1.25	4.03	6.05	617	1.54	4.21	6.28	3	-	-	-
NPL	4	729	0.26	2.16	2.99	617	0.26	2.24	3.10	3	-	-	-
SPL	1	714	0.69	3.44	4.96	602	0.69	3.96	5.49	3	0.69	0.70	1.00
SPL	4	714	0.03	1.81	2.41	602	0.03	1.99	2.66	3	0.38	0.39	0.58
q05	1	656	0.00	1.39	2.78	544	0.02	1.69	3.22	3	0.06	0.08	0.24
q05	4	656	0.00	0.66	1.22	544	0.00	0.81	1.41	3	0.02	0.05	0.14
q10	1	693	0.01	2.03	3.92	581	0.05	2.50	4.49	3	0.13	0.13	0.36
q10	4	693	0.00	1.02	1.83	581	0.01	1.27	2.10	3	0.06	0.07	0.20
q25	1	746	0.26	3.69	6.30	634	0.26	4.36	7.09	3	0.26	0.29	0.62
q25	4	746	0.07	1.94	3.21	634	0.07	2.30	3.62	3	0.14	0.15	0.35
q50	1	828	0.49	6.55	9.92	716	0.49	7.73	10.98	3	0.49	0.57	1.02
q50	4	828	0.18	3.57	5.35	716	0.18	4.06	5.92	3	0.26	0.31	0.60

Note: This table reports the minimum, median, and mean value of consumption floors across surveys using the method proposed by Ravallion (2016) (see equation 1). The floors depend on a poverty line z and a curvature parameter α used for weighting. A linear weight is assumed when $\alpha = 1$, a quadratic weight with higher weight to the bottom values when $\alpha = 2$, and so on. Floors are expressed in daily per capita 2017 PPP\$. Negative income observations are not included. See also Table 3.

Acronyms: z - poverty line; Min - minimum; p50 - median; IPL - International poverty line; NPL - national poverty line; SPL - societal poverty line; q05 - threshold income of 5th percentile; q10 - threshold income of the 10th percentile; q25 - threshold income of the 25th percentile; q50 - threshold income of 50th percentile.

Table D2: Floors using Beta distribution, including negative floors

z	All surveys				Post-2000 surveys				LIC surveys			
	N	Min	p50	Mean	N	Min	p50	Mean	N	Min	p50	Mean
(a) Consumption surveys												
IPL	809	-202.3	0.60	0.70	698	-202.3	0.62	0.39	164	-2.23	0.60	2.28
NPL	809	-73.29	1.05	0.96	698	-73.29	1.12	0.98	164	-0.25	0.65	0.75
SPL	809	-17.85	1.09	1.21	698	-17.85	1.16	1.27	164	-1.14	0.74	0.74
q05	809	-20.83	0.43	0.46	698	-20.83	0.51	0.49	164	-2.26	0.16	0.22
q10	809	-88.64	0.62	0.69	698	-88.64	0.72	0.74	164	-3.07	0.26	0.26
q25	809	-1.54	1.10	1.35	698	-1.54	1.27	1.46	164	-1.20	0.44	0.50
q50	809	-0.16	1.86	2.18	698	-0.16	2.09	2.33	164	-0.16	0.80	0.84
(b) Income surveys												
IPL	1017	-7385	0.11	16.91	905	-7385	0.22	19.08	3	0.22	0.23	0.24
NPL	1017	-3248	0.50	6.11	905	-3248	0.51	6.34	3	-	-	-
SPL	1017	-3453	0.41	16.36	905	-3453	0.48	18.38	3	0.23	0.27	0.32
q05	1017	-5077	-0.06	1.86	905	-5077	-0.01	2.20	3	-0.13	-0.06	-0.05
q10	1017	-2777	-0.01	14.92	905	-2777	0.07	16.81	3	0.00	0.03	0.05
q25	1017	-1894	0.25	19.61	905	-1894	0.35	22.04	3	0.01	0.02	0.10
q50	1017	-1880	1.71	-0.25	905	-1880	1.98	-0.39	3	0.10	0.13	0.36

Note: This table reports the values of the consumption floor using the Beta distribution as outlined in equation 2. Floors are expressed in daily per capita 2017 PPP\$. See also Table 4.

Acronyms: z - poverty line; N - number of surveys; Min - minimum; p50 - median; IPL - International poverty line; NPL - national poverty line; SPL - societal poverty line; q05 - threshold income of 5th percentile; q10 - threshold income of the 10th percentile; q25 - threshold income of the 25th percentile; q50 - threshold income of 50th percentile.

Table D3: Income floors using Three Parameter Log Normal, Dagum III, and LIS approach

Various distributions					(d) LIS-type floors					
Sample	N	Min	p50	Mean	Sample	k	N	Min	p50	Mean
(a) TPLN, LMLE										
All	33	0.08	0.45	0.66	All	2	1009	0.03	2.02	3.92
Post-2000	31	0.08	0.51	0.69		3	1009	0.01	0.87	1.97
						4	1009	<0.01	0.38	1.01
(b) TPLN, ME										
All	870	0.01	7.18	12.45	Post-2000	2	905	0.03	2.91	4.32
Post-2000	798	0.01	7.90	13.33		3	905	0.01	1.32	2.18
LIC	1	0.38	0.38	0.38		4	905	<0.01	0.58	1.12
(c) Dagum III					LIC	2	3	0.04	0.05	0.12
All	29	0.01	0.61	0.75		3	3	0.01	0.01	0.04
Post-2000	28	0.01	0.68	0.77		4	3	<0.01	<0.01	0.01

Note: This table reports the values of the income floors using the Three Parameter Log Normal, Dagum III, and LIS-type approach. LIS floors use the expression: $y_f = \exp(Q_1 - k \times (Q_3 - Q_1))$, where Q_1 is the 1st quartile, Q_3 is the 3rd quartile, and k is the distance parameter. While the table reports the floors using $k \in 2,3,4$, LIS uses $k = 3$. Floors are expressed in daily per capita 2017 PPP\$. See also Tables 5, 6, and 7. Negative floors are excluded.

Acronyms: N - number of surveys; Min - minimum; p50 - median;

Table D4: Floors from various outlier detection methods

z	All surveys				Post-2000 surveys				LIC surveys			
	N	Min	p50	Mean	N	Min	p50	Mean	N	Min	p50	Mean
(a) Consumption surveys												
LIS floor	809	0.01	0.11	0.13	698	0.01	0.12	0.14	164	0.01	0.05	0.05
z-score:												
mean+IQR	809	-6.10	-0.56	-0.68	698	-3.67	-0.44	-0.56	164	-3.67	-1.50	-1.57
mean+MAD	809	-2.84	0.67	0.55	698	-1.76	0.78	0.65	164	-1.76	-0.20	-0.29
mean+Qn	809	-4.26	0.08	-0.06	698	-2.73	0.19	0.04	164	-2.73	-0.86	-0.93
mean+Sn	809	-4.29	0.08	-0.04	698	-2.67	0.20	0.06	164	-2.67	-0.82	-0.91
mean+sd	809	-4.80	0.00	-0.14	698	-2.81	0.13	-0.03	164	-2.81	-1.00	-1.04
median+IQR	809	-2.54	0.64	0.52	698	-1.81	0.76	0.62	164	-1.81	-0.24	-0.33
median+MAD	809	-5.79	-0.59	-0.70	698	-3.70	-0.46	-0.59	164	-3.61	-1.54	-1.61
median+Qn	809	-3.95	0.04	-0.09	698	-2.67	0.16	0.01	164	-2.67	-0.92	-0.97
median+Sn	809	-3.98	0.05	-0.07	698	-2.63	0.18	0.03	164	-2.62	-0.87	-0.95
median+sd	809	-4.49	-0.04	-0.17	698	-2.76	0.11	-0.06	164	-2.76	-1.05	-1.08
(b) Income surveys												
LIS floor	1017	0.02	0.31	0.43	905	0.02	0.35	0.47	3	0.02	0.02	0.04
z-score:												
mean+IQR	1017	-4.44	0.22	0.14	905	-4.44	0.66	0.37	3	-4.08	-3.71	-3.20
mean+MAD	1017	-1.80	1.54	1.52	905	-1.80	1.83	1.69	3	-1.78	-1.72	-1.21
mean+Qn	1017	-3.09	0.84	0.79	905	-3.09	1.17	0.99	3	-2.98	-2.93	-2.31
mean+Sn	1017	-3.05	0.90	0.84	905	-3.05	1.25	1.04	3	-2.89	-2.76	-2.21
mean+sd	1017	-3.21	0.65	0.62	905	-3.21	0.98	0.82	3	-3.21	-3.18	-2.53
median+IQR	1017	-1.76	1.57	1.53	905	-1.76	1.86	1.70	3	-1.76	-1.69	-1.21
median+MAD	1017	-4.32	0.26	0.15	905	-4.32	0.67	0.38	3	-4.06	-3.69	-3.20
median+Qn	1017	-2.97	0.87	0.80	905	-2.97	1.21	1.01	3	-2.96	-2.91	-2.31
median+Sn	1017	-2.93	0.94	0.85	905	-2.93	1.28	1.06	3	-2.87	-2.74	-2.21
median+sd	1017	-3.19	0.68	0.63	905	-3.19	1.00	0.83	3	-3.19	-3.16	-2.53

Note: 1. LIS floor: $\text{EXP} [\log Q1 - 3 * (\log Q3 - \log Q1)]$ 2. The "z-score/(x + y)" method follows Belotti, Mancini, Vecchi (2024) to identify the lower boundary (floor). Here, "x" is the location parameter (mean or median), and "y" is a robust scale estimator for the transformed variable t, as defined in Table 2 of Belotti (2024). Floors are expressed in daily per capita 2021PPP.

Table D5: Share of population below \$0.25 and \$0.50 per day, 10 most affected surveys

Rank	Below \$0.25/day						Below \$0.50/day					
	Consumption			Income			Consumption			Income		
	Country	Year	Pop share, %	Country	Year	Pop share, %	Country	Year	Pop share, %	Country	Year	Pop share, %
1	Zambia	1991	15.93	Suriname	1999	11.04	Congo, Dem. Rep.	2005	30.22	Haiti	2001	16.54
2	Lesotho	1994	7.14	Panama	1989	9.79	Zambia	1991	25.39	Haiti	2012	15.11
3	Lesotho	2003	6.14	Colombia	2000	7.99	Mali	1994	24.04	Micronesia, Fed. Sts.	2000	13.05
4	Congo, Dem. Rep.	2005	6.12	Haiti	2001	7.74	Mozambique	1996	22.33	Panama	1989	11.87
5	Mozambique	1996	4.76	Panama	1991	6.98	Lesotho	1994	16.59	Suriname	1999	11.81
6	Mali	1994	4.68	Haiti	2012	6.95	Niger	1994	16.42	Bolivia	2000	10.20
7	Uzbekistan	1998	4.29	Panama	1995	6.70	Lesotho	2003	16.27	Panama	1991	8.99
8	Gambia, The	1998	3.57	Micronesia, Fed. Sts.	2000	5.82	Burkina Faso	1994	12.92	Colombia	2000	8.53
9	Burundi	1999	3.42	Panama	1998	5.30	Gambia, The	1998	12.53	Bolivia	1999	8.40
10	Benin	2015	3.00	Colombia	1999	5.20	Burundi	1999	12.41	Nicaragua	1993	8.38

Note: This table reports the top 10 income and consumption surveys and the share of population that are below \$0.25 per day and \$0.50 per day thresholds from the most affected (rank 1) to the least.